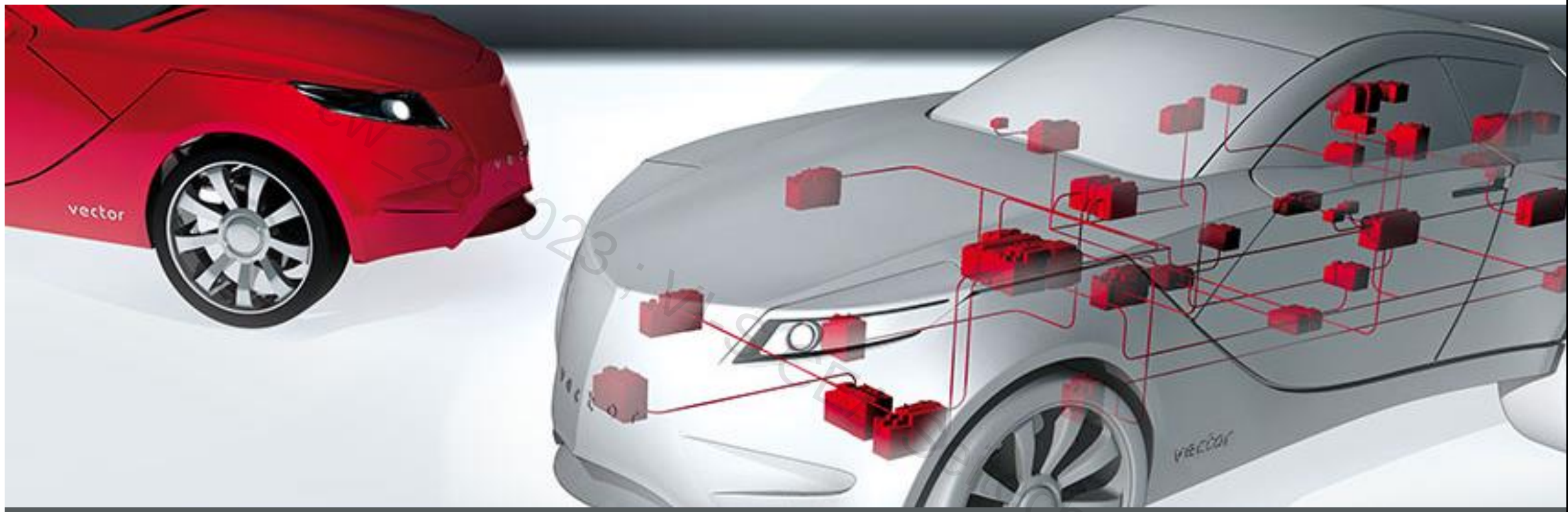
VECTOR



AUTOSAR in Practice

Communication

Agenda

- **Communication**
  - Transmission
  - Signal Groups
  - Update Bit
  - Reception
  - Data Mapping
  - Notification Mechanisms
  - Deadline Monitoring
  - Invalidation
  - Rx Filter
  - Additional Return Values
  - Coming up next

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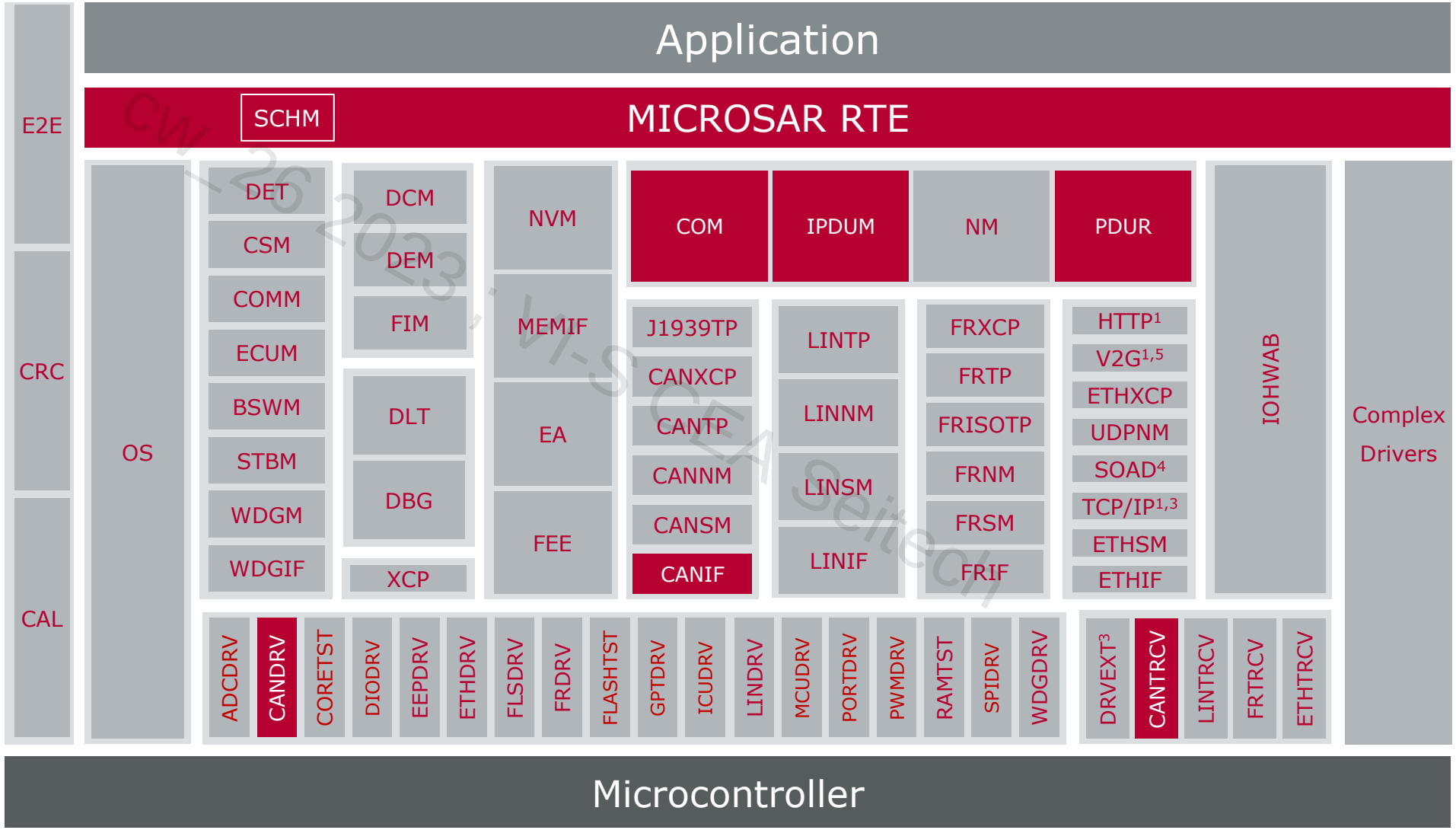
VECTOR

Related modules

- COM
- IPDUM
- PDUR
- CANIF
- CAN
- CANTRCV

- DET
- SCHM



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How do communication modules interact

COM

- Signal interface to RTE
- Signal gateway
- I-PDU transmission and reception
  - Periodic, on event, etc.
- Deadline monitoring
- Notifications to RTE
- Signal Invalidation

PDU Router

- Abstraction of the bus system
- Interface Layer I-PDU gateway
- TP Layer I-PDU gateway

CAN Interface

- HW independent
  - Queuing
  - Handling of CAN controller states

CAN Driver

- HW abstraction
  - Controller initialization
  - ISR implementation
  - Writing/reading of L-PDU data from/to hardware
  - Wake-up and bus-off handling

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COM – Communication
PDUR – PDU Router
CANIF – CAN Interface
CAN – CAN Driver
DET – Development Error Tracer
SCHM – Schedule Manager

How do communication modules interact

COM:

Com operates on I-PDUs and signals, but has no awareness of the underlying bus technology. In AUTOSAR 4.x additional features have been added:

- large data types with max PDU size 4095 Mbytes. Data type UINT8_N.
- dynamic length signals and I-PDUs using the TP (varying size during runtime, max length defined at configuration time, data type UINT8_DYN)

NOT supported in MICROSAR: float data type, data sequence (ComIPduCounter, discarding and notification of violations over ComIPduCounterErrorNotification), communication protection (fan-in with voting)

PDU Router:

The PDU Router is responsible for routing (including multi-cast routing) of messages to the same type of bus system (CAN – CAN) or different bus systems (e.g. CAN – FlexRay) and between service layer modules (Com/Dcm) and transport protocol modules.

CAN interface:

Handles all CAN channels in the system.

CAN Driver:

- Wakeup is reported to EcuM

- Bus-off state is reported to CanIf which forwards the state to CanSm

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How do communication modules interact

CANTRCV

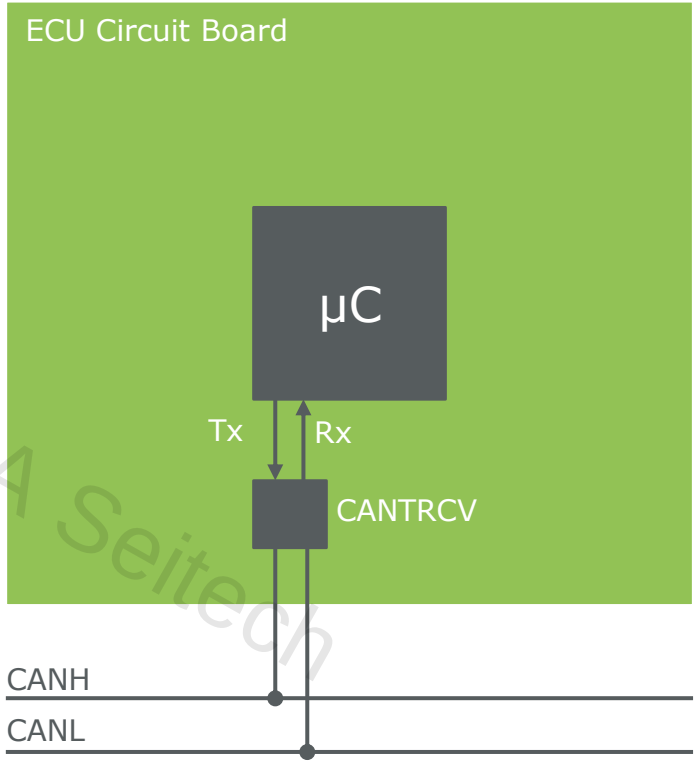
- Convert unidirectional signal µC pins into differential signals on the CAN bus
  - RxD, TxD ←→ CAN_H, CAN_L
- Bus biasing, bus driver control, line error detection, voltage control, wakeup detection
- Selective wakeup for partial networking (including asynchronous transceiver handling)

DET

- Collects all detected errors of the BSW modules
- For development only

SCHM

- Embeds BSW module implementations into AUTOSAR OS context
- Triggers main processing functions of the BSW modules
- Offers data consistency mechanisms for the BSW modules



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The RTE generator is responsible for the creation of the SCHM code. Similarly to how the RTE generates code to execute SWC Runnable Entities, it can do the same thing for the Basic Software Schedulable Entities (main functions). The RTE schedules them according to the internal behavior (IB) of the BSW Modules.

Optional module

IPDUM

- Multiplexed I-PDUs
  - Handle Tx/Rx I-PDU with multiple different signal layouts
  - Payload: zero or more signals or signal groups
  - Static and multiplexed part both contain payload and are flexibly segmented
  - Multiplexed part contains selector for multiplexed layout
- Usually used for CAN
  - LIN and FR also conceivable

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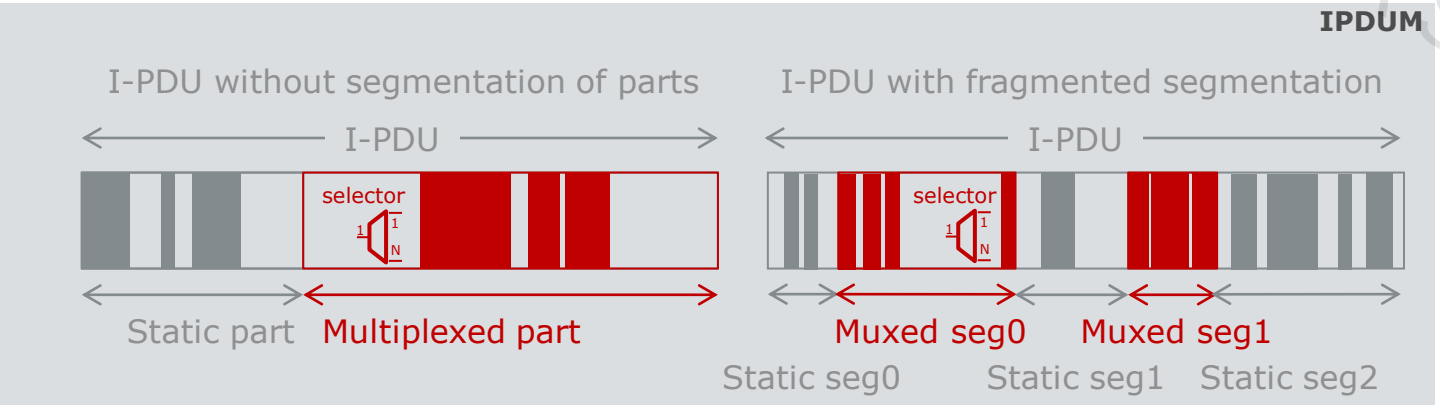
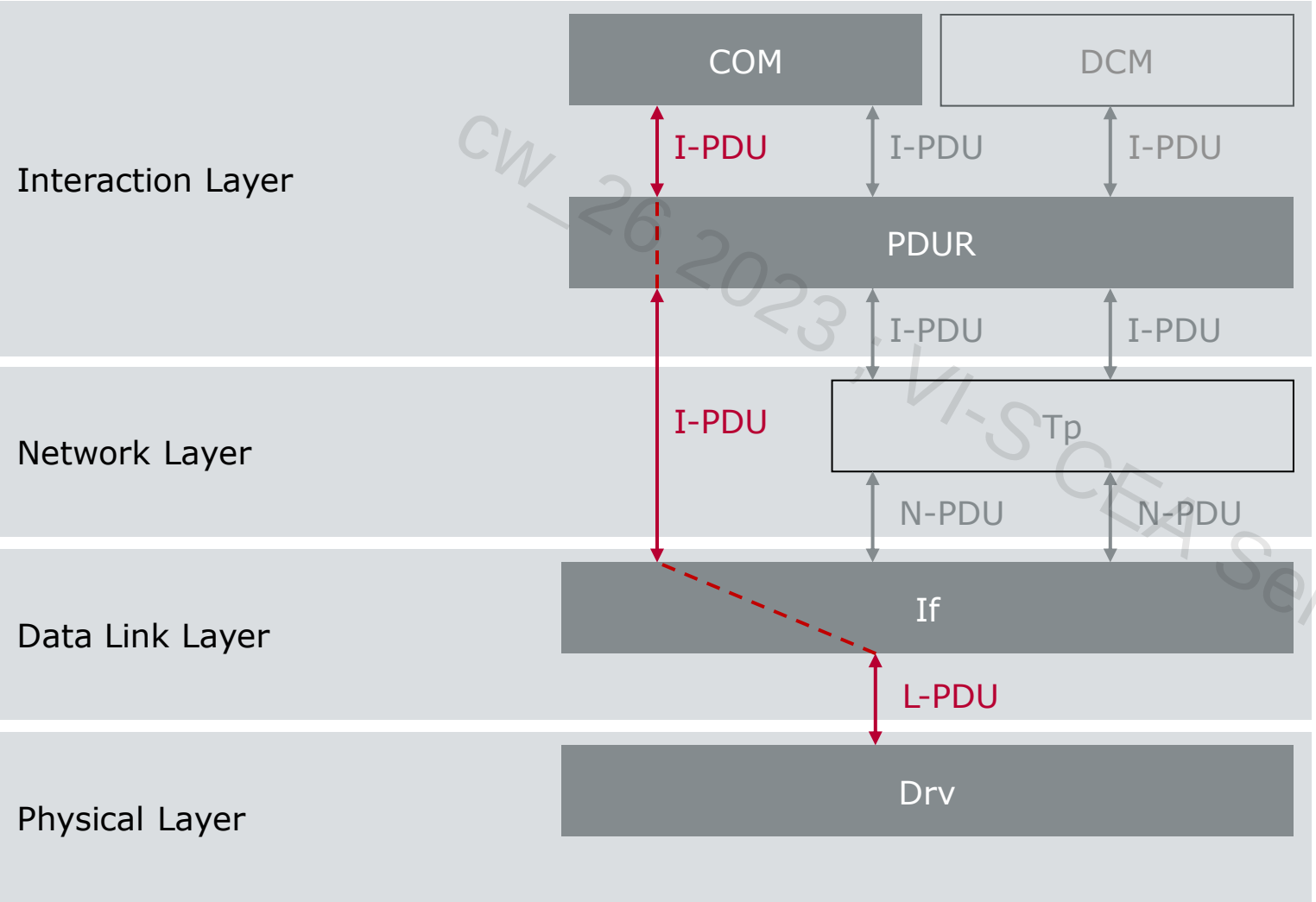


PDU concepts

PDU concept



L-PDU: Link Layer Protocol Data Unit

N-PDU: Network Layer Protocol Data Unit

I-PDU: Interaction Layer Protocol Data Unit

DCM: Diagnostic Communication Manager (introduced in chapter "Diagnostics")

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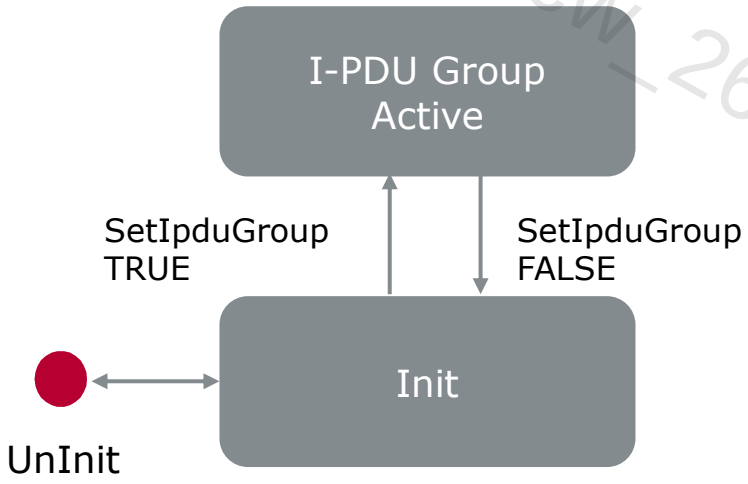
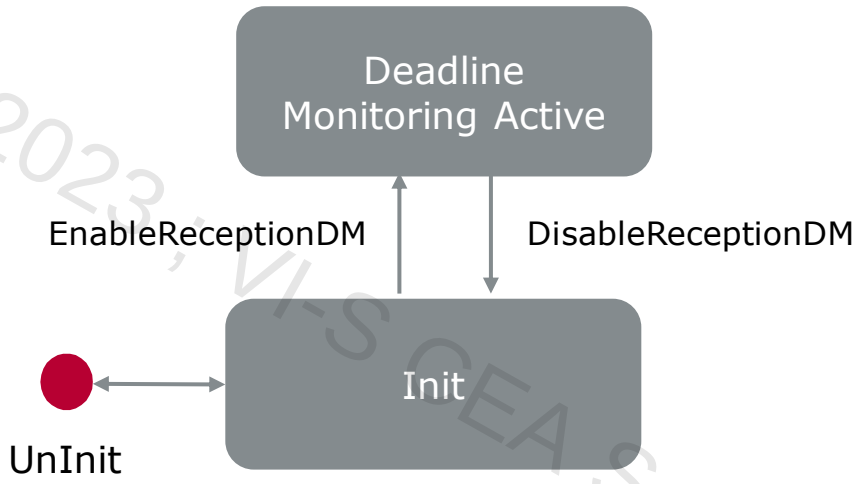
COM I-PDU Groups

- States of I-PDU Groups





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- Controlling of
  - Transmission path
  - Reception path

- Controlling of
  - Deadline monitoring

COM I-PDU Groups

-I-PDU Groups allow grouping of I-PDUs to channel-specific groups. The Bus State Managers (e.g. CanSm) switch the corresponding channel states and notify the BswM. The BswM uses the channel specific groups to control the bus communication based on these notifications.

-The EcuM initializes the Com and allows to perform the transition in the above state diagrams from “UnInit” to “Init” state.

-The activation and deactivation of I-PDU Groups and Deadline Monitoring are handled independently, as shown in the two state machines.

-In AUTOSAR 4, it is up to the **BswM** to set the I-PDU Groups and to control them. For this the APIs `Com_SetIpduGroup()` and `Com_IpduGroupControl()` are used.

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Agenda

- Communication
- **Transmission**
- Signal Groups
- Update Bit
- Reception
- Data Mapping
- Notification Mechanisms
- Deadline Monitoring
- Invalidation
- Rx Filter
- Additional Return Values
- Coming up next

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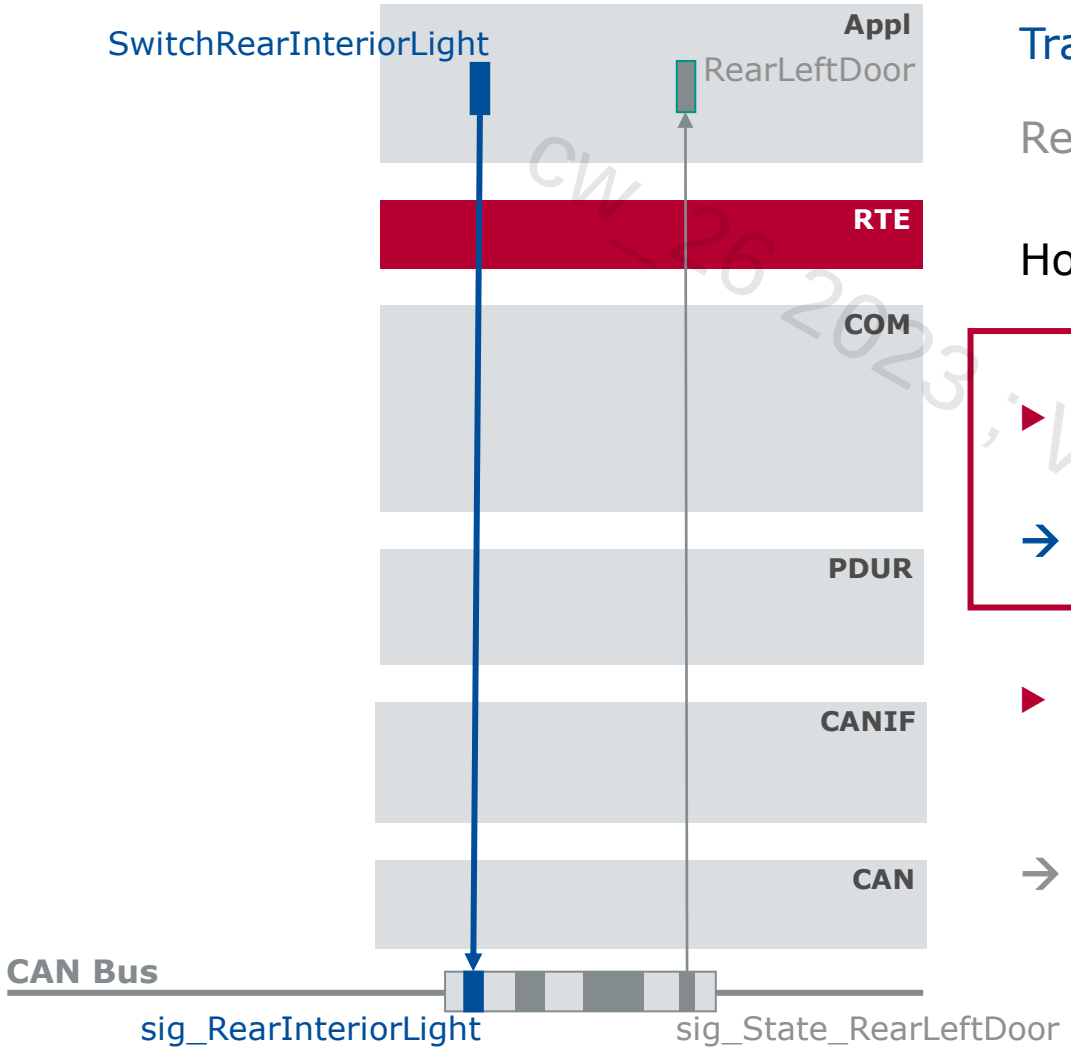
Transmission and Reception



Transmission?

Reception?

How does it work?

- Follow a data element on its way down to the bus

→ **transmission**

- Follow a bus signal on its way up to become a data element

→ **reception**

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Signal transmission: Appl, RTE, COM

RTE Direct / Explicit Write

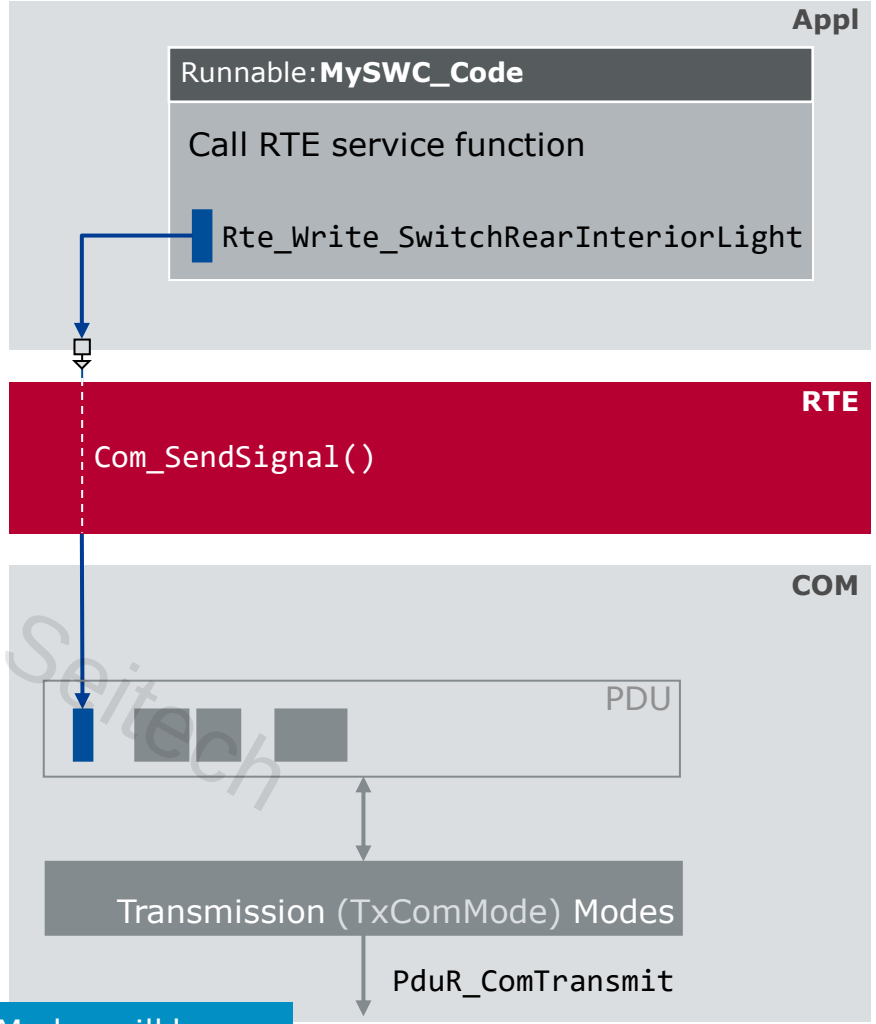
- The information is **immediately** copied to buffer in Com.

RTE Buffered / Implicit Write

- The information is copied to buffer in Com **after** runnable is finished.

Com Transmission Modes

- Two Tx modes per I-PDU
- Which mode is used is based on signal filters



Transmission Modes will be discussed later

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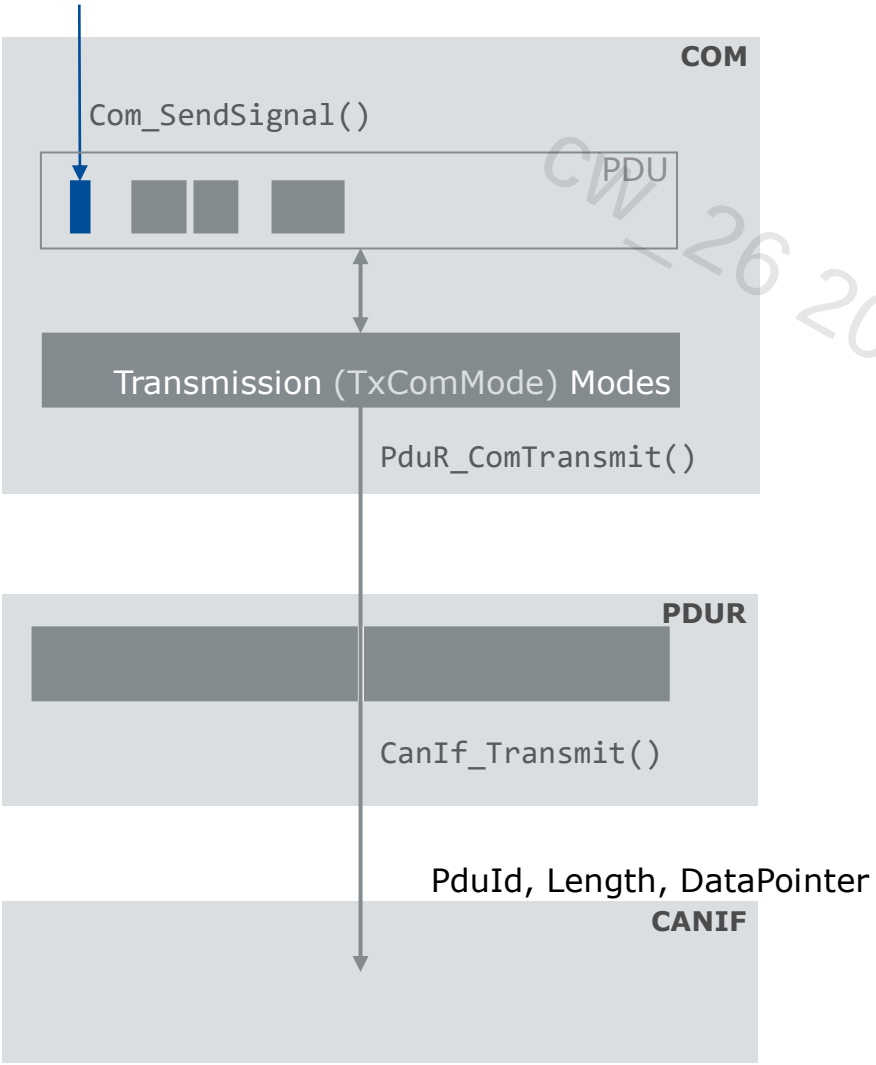
Signal transmission: COM, PDUR, CANIF



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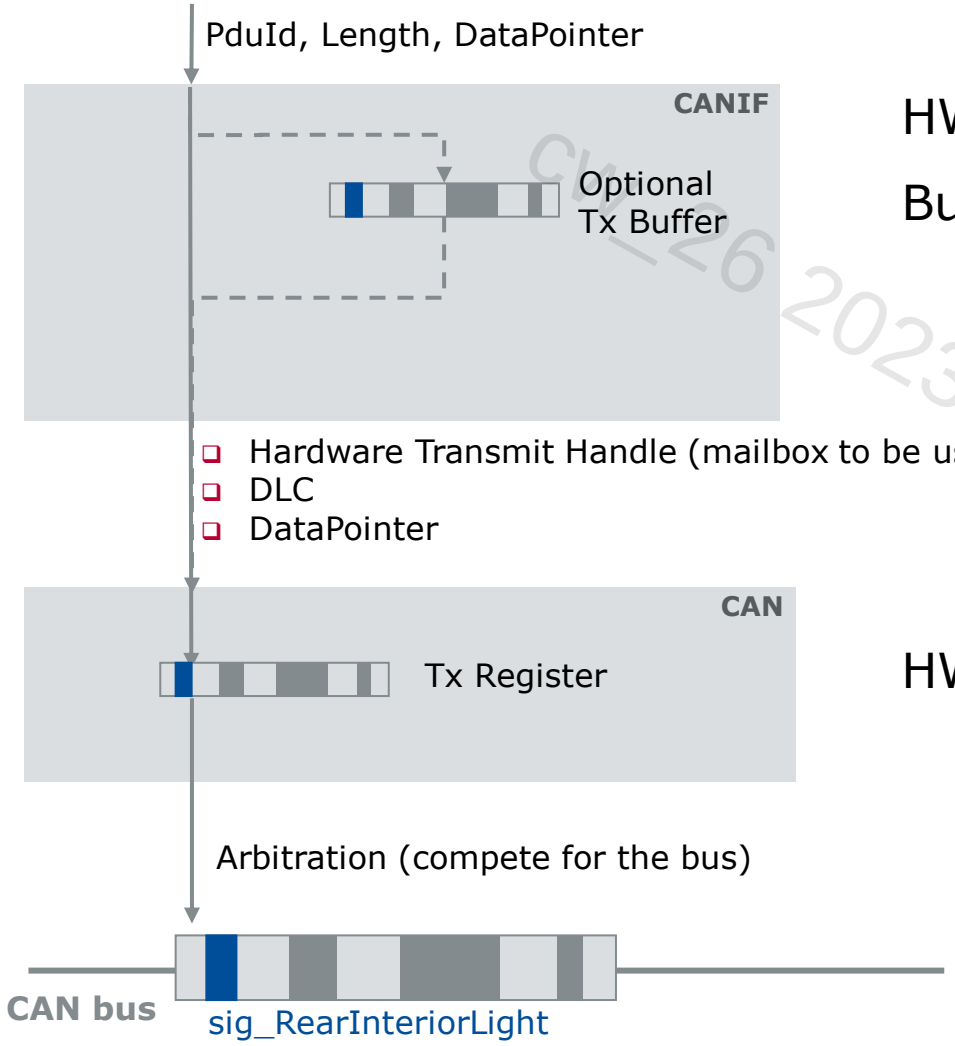
Signal transmission: CANIF, CAN, Bus



HW-independent

Buffer for transmit data

HW-dependent

A Tx Buffer on CanIf level is not required if FullCAN is used or ComRetryFailedTransmitRequests == TRUE.

ComRetryFailedTransmitRequests is new in AUTOSAR 4.x. However in Vector Com it is already implemented since version 3.x.

In the case where ComRetryFailedTransmitRequests == TRUE, the Com evaluates the return value of PduR_ComTransmit and in case of a rejected transmission, the Com will call PduR_ComTransmit during the next Com_MainfunctionTx call. Only the repeated transmission attempt can be delayed; neither Tx Mode nor regular repetitions are affected by the retries.

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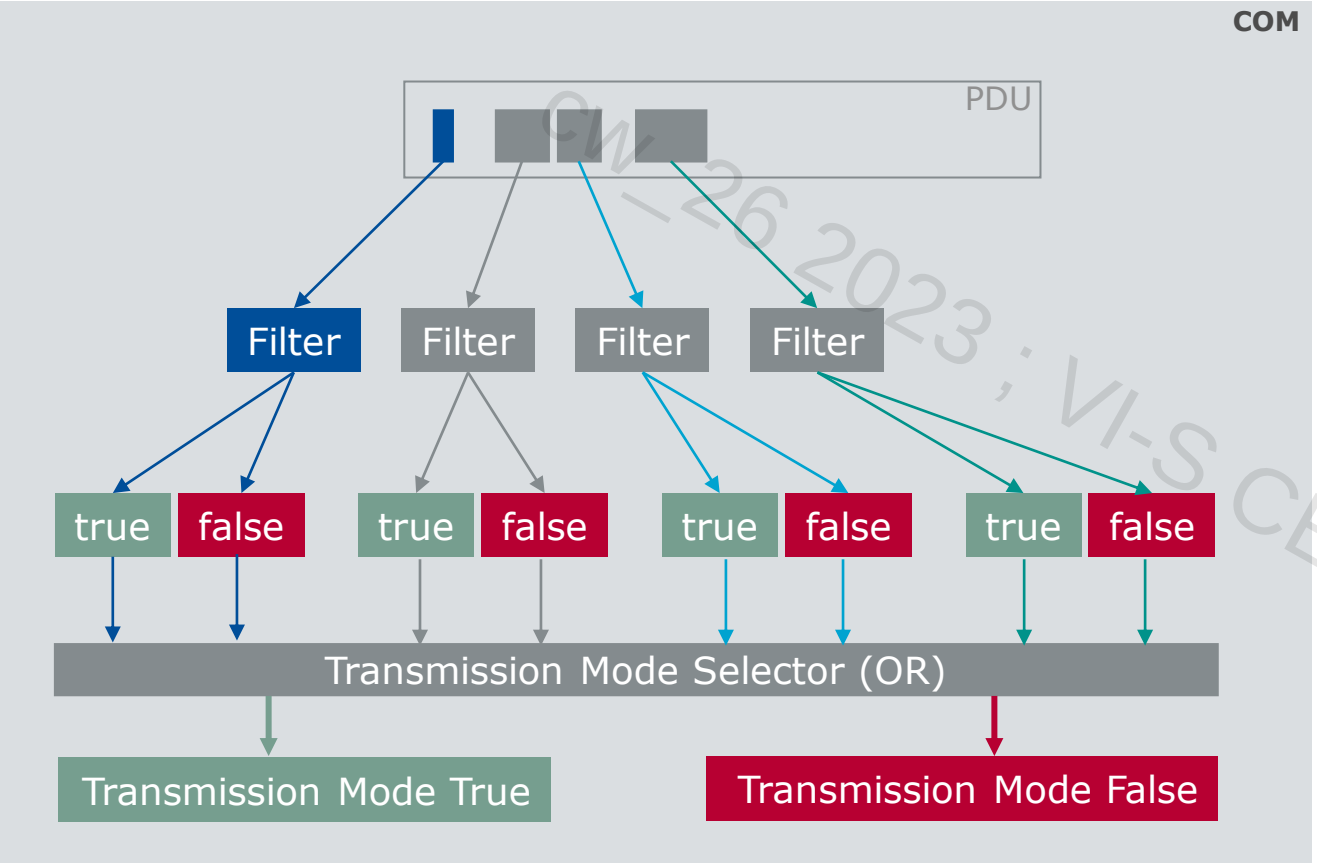
Transmission Modes - Mechanism



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What about the **FILTERS**?

In the above picture, only TX filters are shown. Nevertheless there are also RX filters (see next slides)

Transmission Modes – COM Signal Filters

Filter

Available signal-oriented Filters - Part 1

- **None:**
  The signal has no filter

- **Always**:
  This filter always evaluates to TRUE
  → No Tx mode "False" required

- **Never**:
  This filter always evaluates to FALSE

- **MaskedNewDiffersMaskedOld**:
  ((new & mask) != (old & mask))

oldValue:0000 1010 newValue:1011 1001

mask = 0000 1111 → true
mask = 0000 1100 → false

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Transmission Modes – COM Signal Filters

Filter

Available signal-oriented Filters Part 2

- **MaskedNewEqualsX**:
  ((new & mask) == x)
  newValue:1011 1001 X=9
  mask = 0000 1111 → true
  mask = 1001 0000 → false
- **MaskedNewDiffersX**:
  ((new & mask) != x)
  newValue:1011 1001 X=9
  mask = 0000 1111 → false
  mask = 1001 0000 → true
- **NewIsOutside**:
  ((new < min) || (max < new))
  min = 10, max = 103
  newValue = 10 → false
  newValue = 254 → true
- **NewIsWithin**:
  ((min <= new) && (new <= max))
  min = 10, max = 103
  newValue = 10 → true
  newValue = 254 → false

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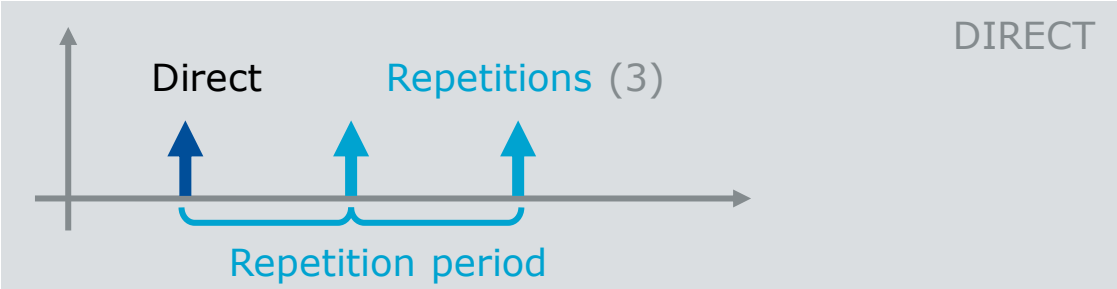
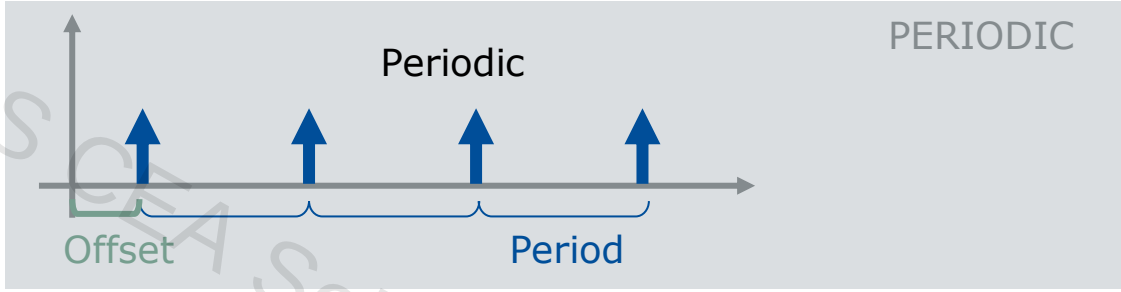
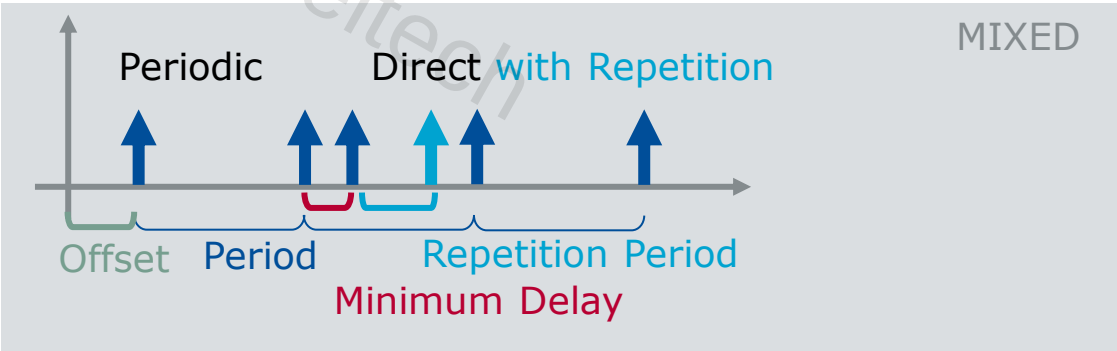
Transmission Modes - I-PDU Tx Modes

Available I-PDU Transmission Mode values

- **NONE**
- **DIRECT**
  - Repetition
  - Repetition Period
- **PERIODIC**
  - Period
  - Offset
- **MIXED**
  - Direct and Periodic
  - Period
  - Offset
  - Repetition
  - Repetition Period
  - Minimum Delay







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Transmission Modes - I-PDU Tx Modes

AUTOSAR PARAMETER DEFINITION: ComTxModeTrue, ComTxModeFalse

The **Transmission Mode** determines the Tx properties of the I-PDU:

- **DIRECT**: the message is transmitted in the next `Com_MainFunctionTx()` call after writing to a signal.
- **PERIODIC:** the I-PDU is transmitted at the specified cycle time (ComTxModeTimePeriodFactor).
- **MIXED**: Combines the transmission modes PERIODIC and DIRECT.
- **NONE**: Com does not transmit these I-PDUs. Only lower layers can request the Com to transmit an I-PDU with transmission mode NONE by calling the function `Com_TriggerTransmit()`

Generally, the Com triggers the transmission of any I-PDU by calling `PduR_ComTransmit()`.

All transmission modes, except NONE, maintain the Minimum Delay Time (ComTxIPduMinimumDelayTimeFactor) and delay a transmission to meet this time if required.

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Transmission Modes - I-PDU Tx Modes

Switching from Transmission Mode “True” to Mode “False”

- All signal filters of an I-PDU are evaluated
  - Mode “True” is used if any filter evaluates to True
  - Mode “False” is used if all filters evaluate to False

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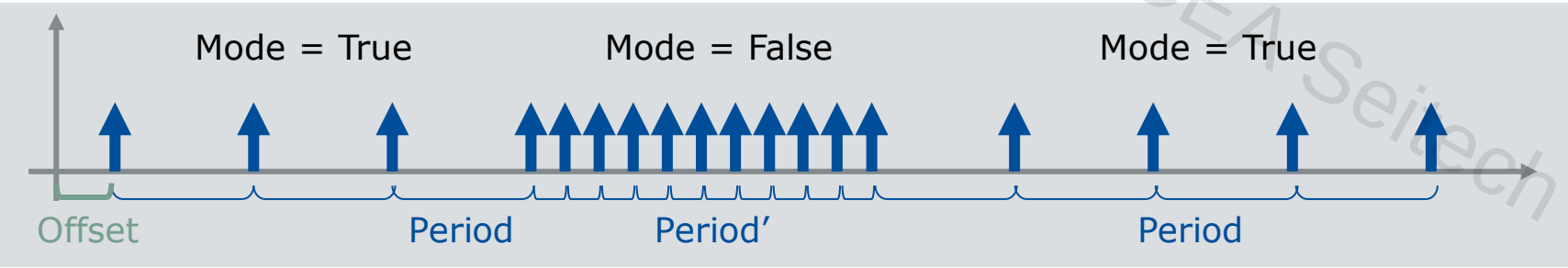
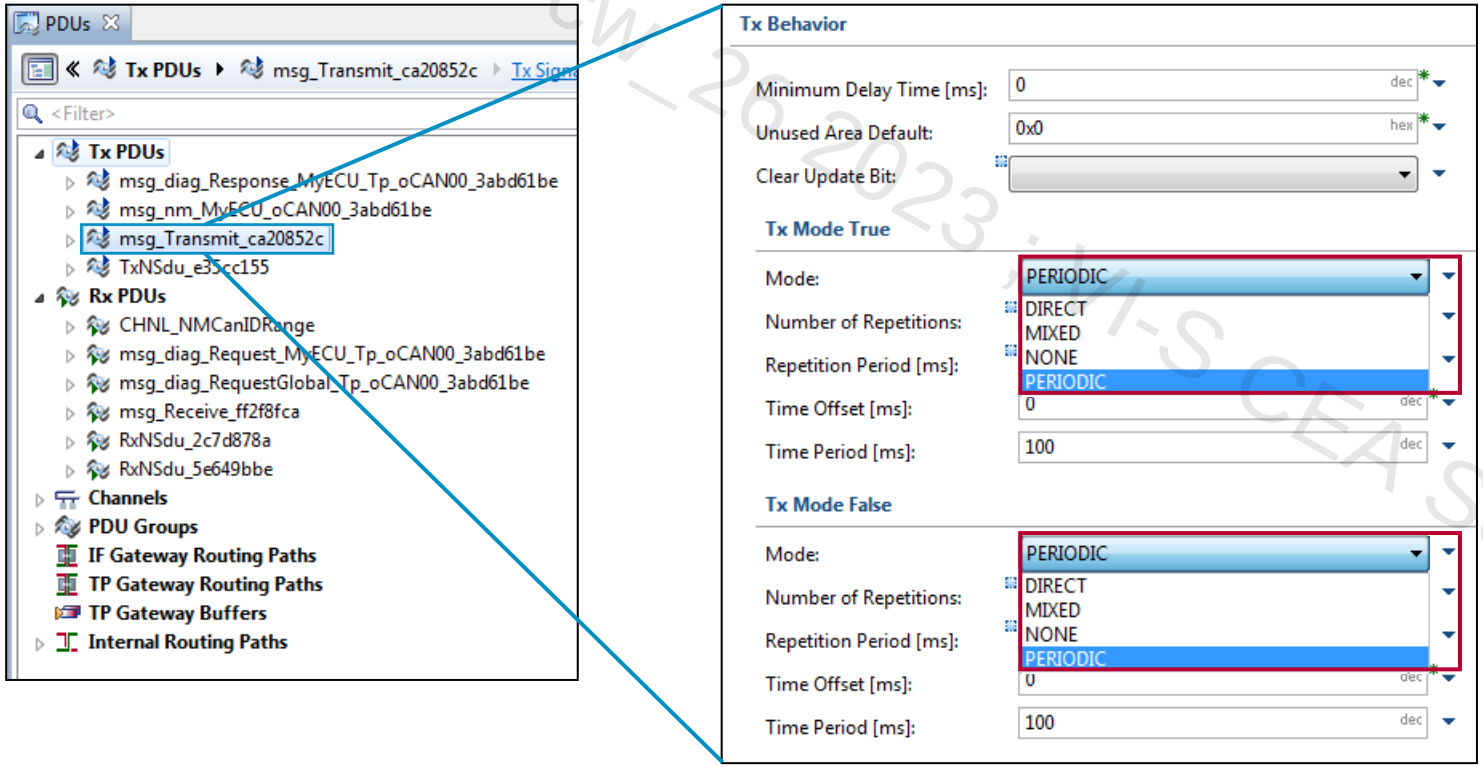


Transmission Modes in Configurator

The COM Transmission Modes are
I-PDU Specific settings



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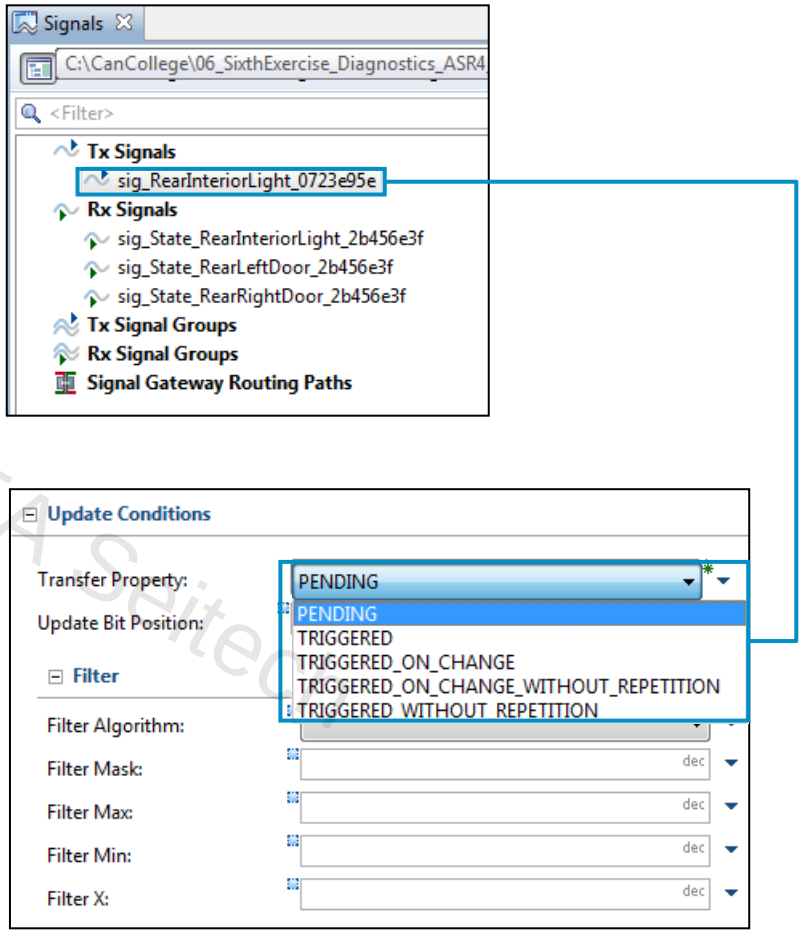
Transfer Property in Configurator

The COM Transfer Property is Signal-Specific

- **PENDING**
  writing data to this signal will not trigger a direct transmission
- **TRIGGERED** (see notes)
  writing data to this signal is the trigger for a transmission
  - For Direct or Mixed I-PDU Transmission Modes
  - On Write or On Change (with Repetition)
  - On Write or On Change without Repetition



For a Transmission Mode DIRECT or MIXED the Transfer Property must be set to TRIGGERED.

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Transfer Property in Configurator

The Transfer Property is only used in connection with the Direct or the Mixed Transmission Mode. Exception

Set to **TRIGGERED** means: writing the signal is a trigger for direct transmission of the corresponding I-PDU

Set to **PENDING** means: even writing the signal will not trigger direct transmission of the corresponding I-PDU.

Example: all signals within a PDU are set to **Pending**. Transmission Mode is set to **Direct**. The writing any of the signals will not provoke a transmission of the PDU.

TRIGGERED - cause a transmission with repetitions according to the Tx Mode when signal is written

TRIGGERED_WITHOUT_REPETITION - ignore repetitions of Tx mode and transmit only once

TRIGGERED_ON_CHANGE - trigger transmission including repetitions if signal differs from the locally stored value (last sent or init).

TRIGGERED_ON_CHANGE_WITHOUT_REPETITION - trigger transmission without repetitions if signal differs to the locally stored (last sent or init) value.

The AUTOSAR Com module shall not support the transfer properties TRIGGERED_ON_CHANGE and TRIGGERED_ON_CHANGE_WITHOUT_REPETITION for signals and group signals with ComBitSize configured to 0.

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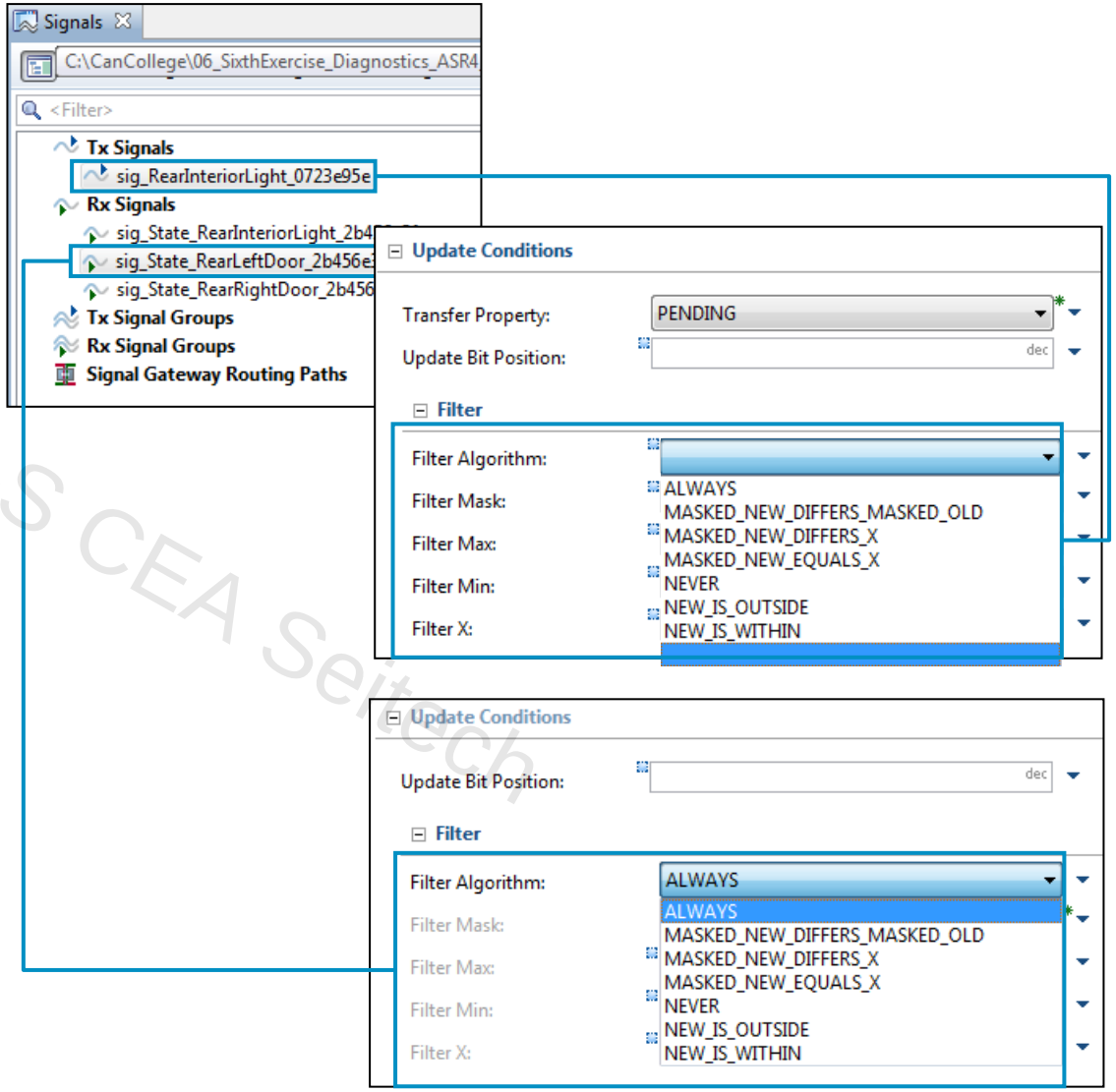
Signal Filters in Configurator

- The COM Filter settings are Signal-Specific
- Signal filters can be defined for Rx or Tx signals
- Signal filters have an impact on
  - The transmission mode selector (Tx)
  - On the indication of the respective signals (Rx)



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Signal Filters in Configurator

For the transmission path, there are various filter settings. Filters are optional and in some situations the field can be left empty. If this is the case, the respective signal content is not considered for the Transmission Mode selection.

For the reception path, there is also the possibility to filter the reception seen by the application. The easiest (and default) behavior can be characterized as “ALWAYS” where there is no checking in the Com for signal values compared to the filters. However this can increase the processing time in the application. Therefore you can also select to be notified only on change by “MASKED_NEW_DIFFERS_MASKED_OLD” to decrease the triggering rate.

Since AUTOSAR 4.1.1 , there is one more filter called ONE_EVERY_N. This filter means Com counts transmission requests and on the given value, the transmission is carried out.

If the ComFilterAlgorithm ONE_EVERY_N is used, the value of the signal itself has no influence on the Transmission Mode selection. What counts is the number of send requests by the RTE.

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Agenda

- Communication
- Transmission
- **Signal Groups**
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- Additional Return Values
- Coming up next

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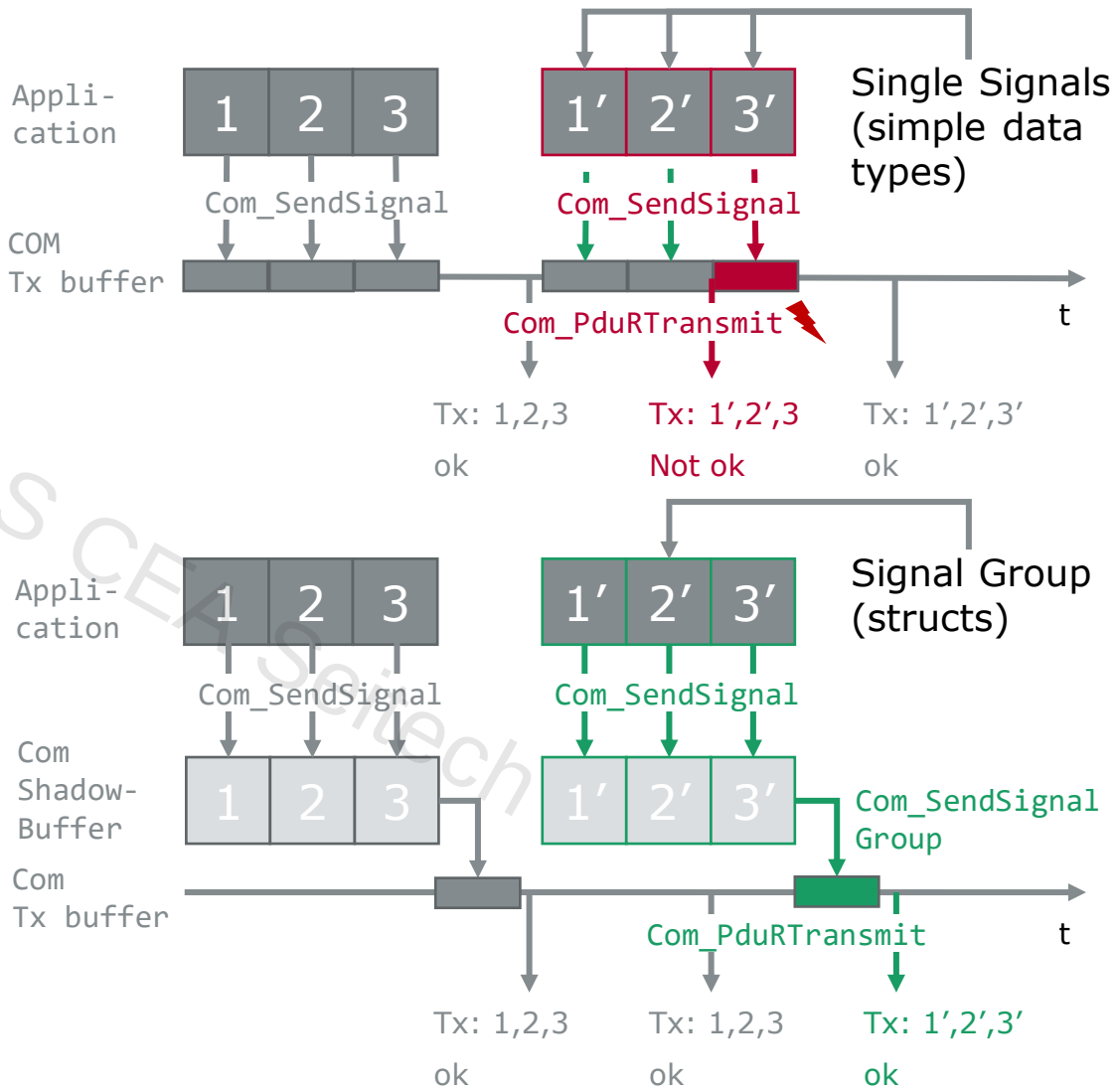
Data consistency using signal groups

Signal Groups

- Logical group of signals that belong together
- Signal groups are defined in the communication matrix
- Complex data types are mapped to signal groups by the RTE
- Signal groups comprise group signals

Signal Groups signify the following

- All group signals within a signal group are consistent
- Access is done by the RTE via a dedicated signal group and group signal API
- Signal group shadow buffer provided by Com
  - One global shadow buffer for each signal group



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Data consistency using signal groups

Example: data that may be grouped in a signal group are the signals for velocity and vehicle direction

In AUTOSAR 4.x the API `Com_SendSignal()` writes a group signal into the shadow buffer, not directly to the I-PDU.

After all group signals of a signal group have been updated, the consistent content of the shadow buffer is "transmitted" using the API `Com_SendSignalGroup()`. A similar API is available for signal group reception.

Note: In AUTOSAR versions before 4.x, the API `Com_UpdateShadowSignal()` is used instead of `Com_SendSignal()` to update Group signals. This API is now deprecated.

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Agenda

- Communication
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- ▶ **Update Bit**
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Concept

Update Bit

- Is defined within the communication matrix
- Allows the receiver to determine if the transmitter has updated the value

What is the Update Bit?

- Dedicated bit within the I-PDU of the related signal
- Not directly accessible for the application or RTE
- The update bit is **set by COM** when signal is written
- Update bits are **cleared by COM** according to
  - *ComTxIPduClearUpdateBit = {Confirmation|Transmit|TriggerTransmit}*
- Usage of update bit causes signal-based handling of
  - Notification functions
  - Deadline monitoring

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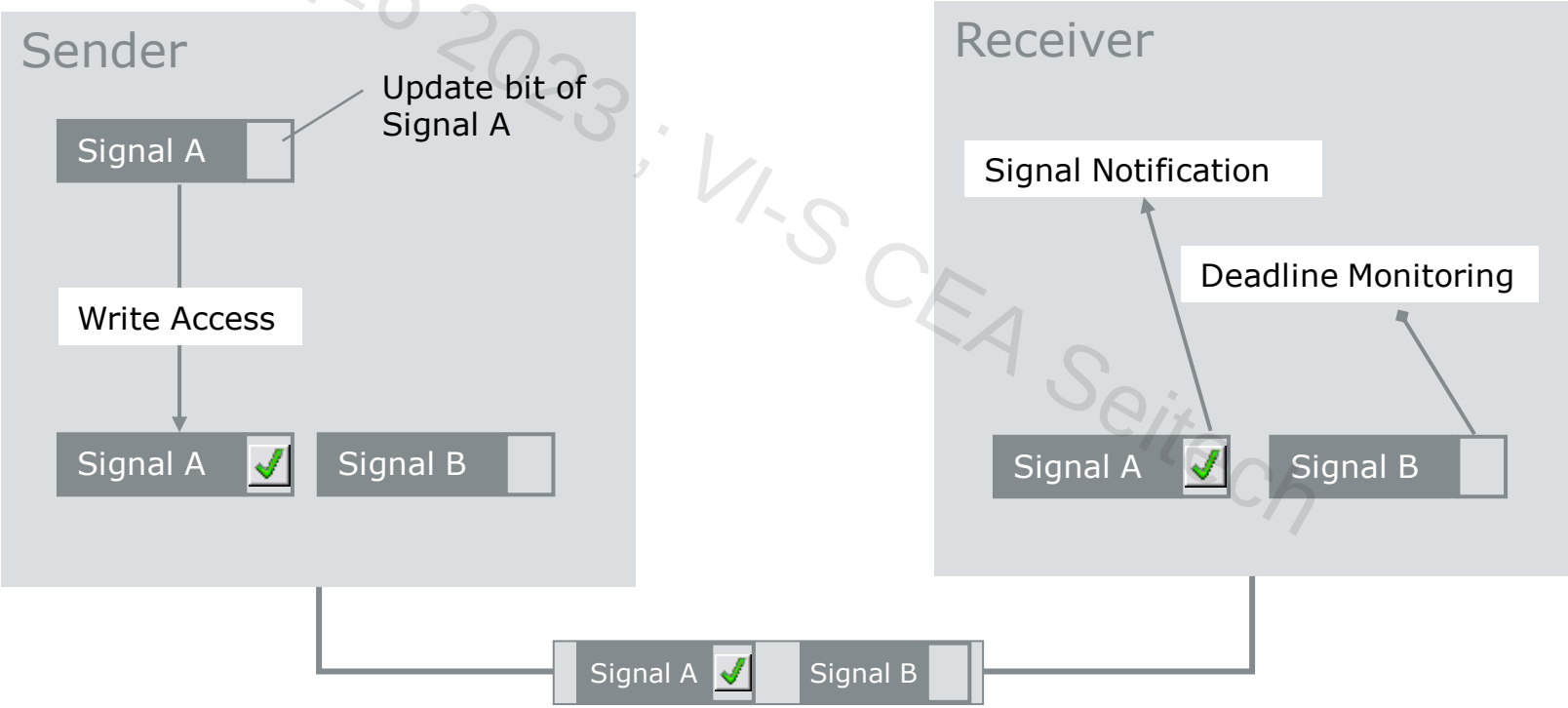
The update bit is set independently of the signal filter result

Concept

- Update bits allow a signal-based timeout observation and a signal-based receive notification



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Agenda

- Communication
- Transmission
- Signal Groups
- Update Bit
- ▶ **Reception**
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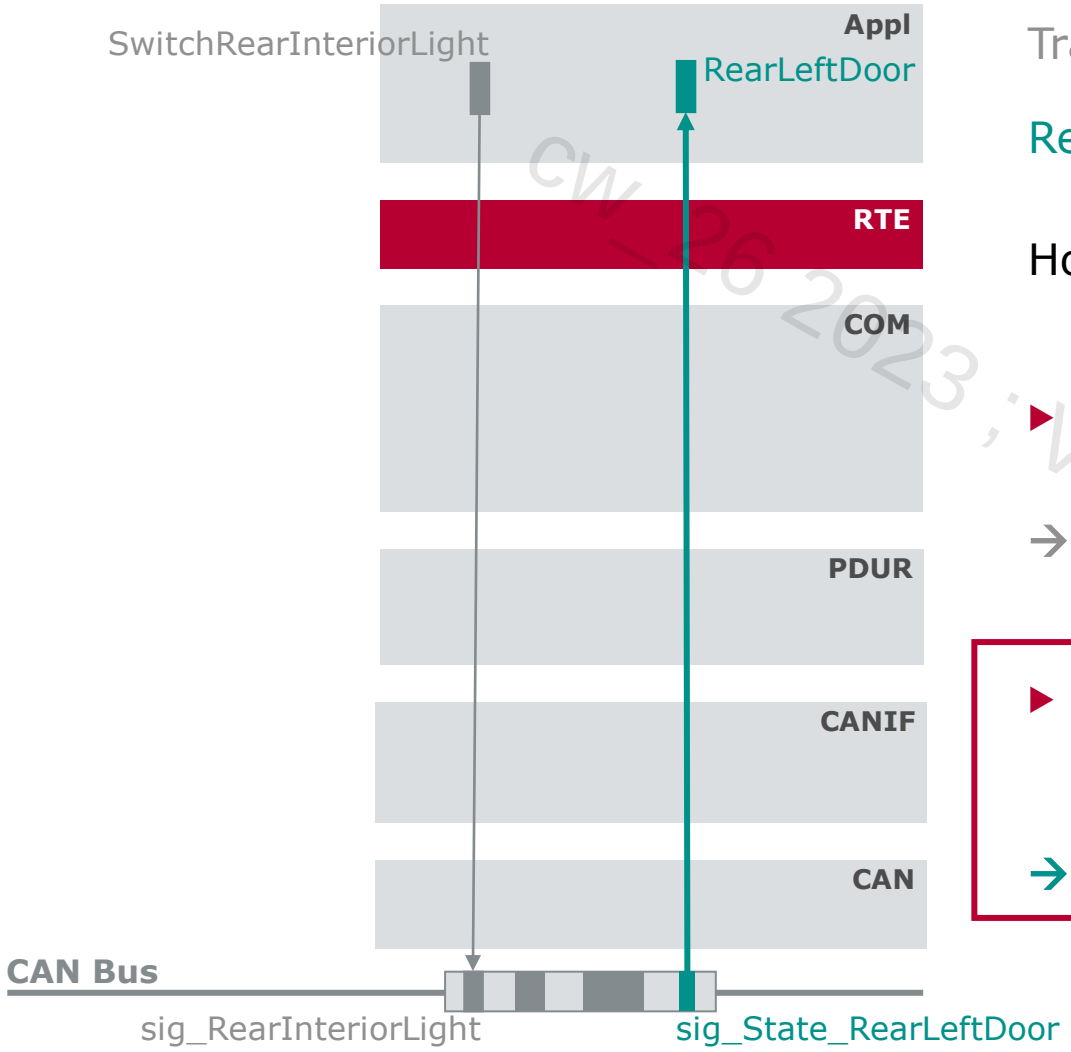
Signal Reception



Transmission?

Reception?

How does it work?

- Follow a data element on its way down to the bus

→ **transmission**

- Follow a bus signal on its way up to be used as a data element

→ **reception**

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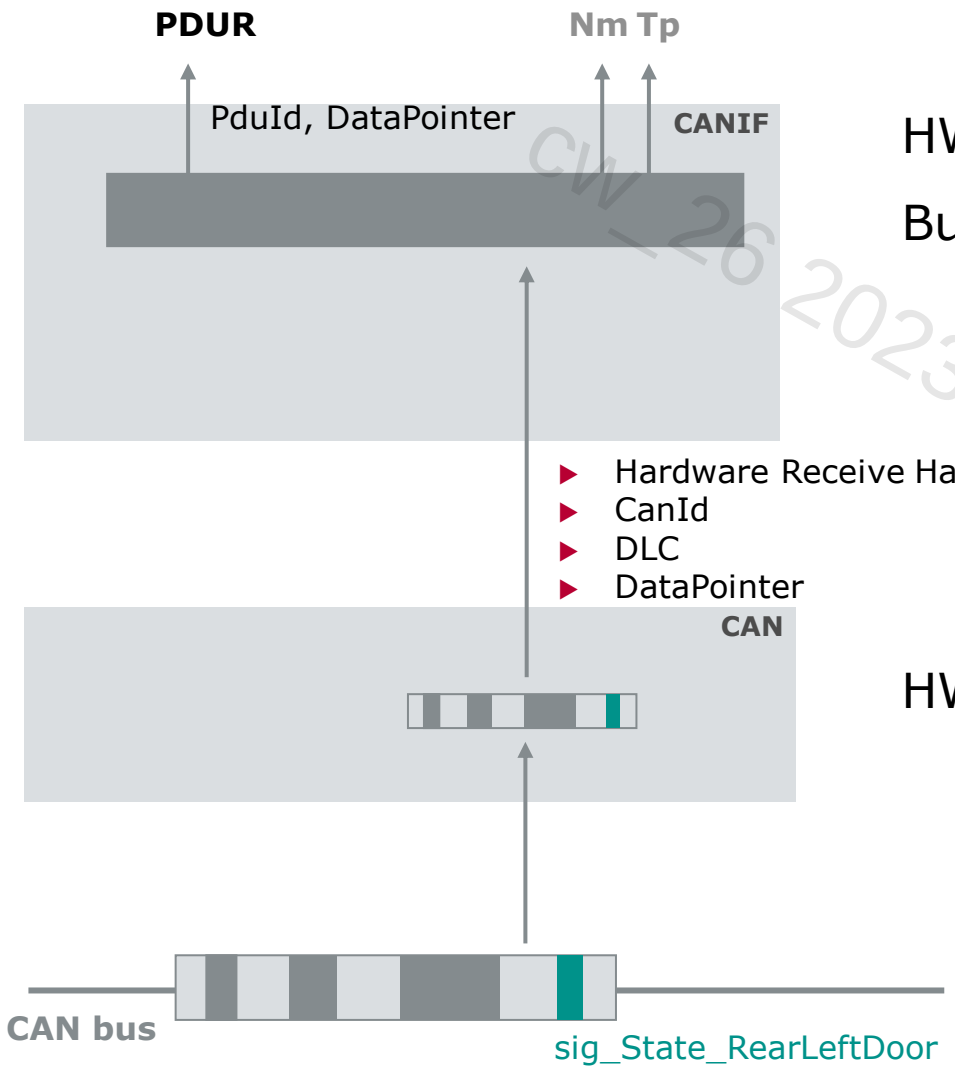
Signal Reception: Bus, CAN, CANIF



HW independent

Buffer for receive data

HW dependent

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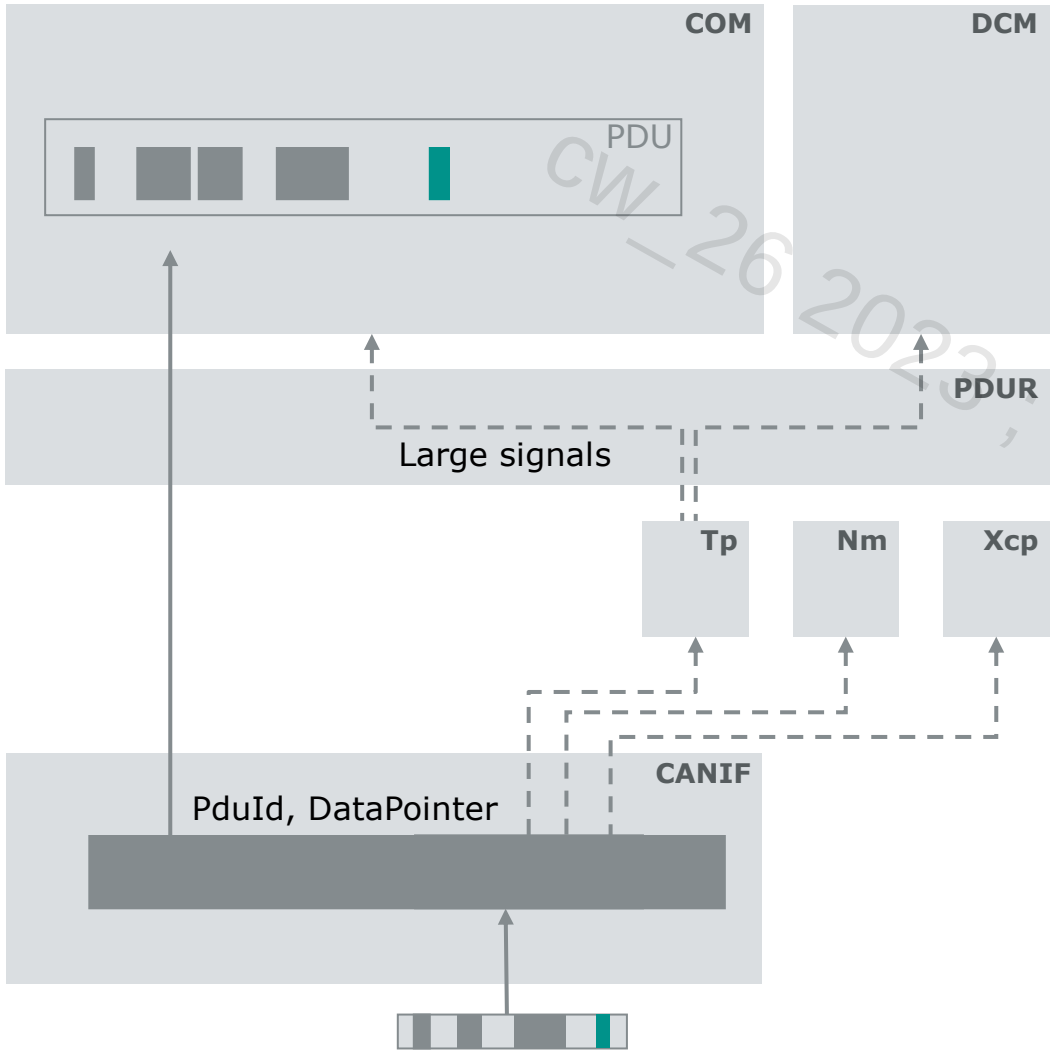
Signal Reception: CANIF, PDUR, COM



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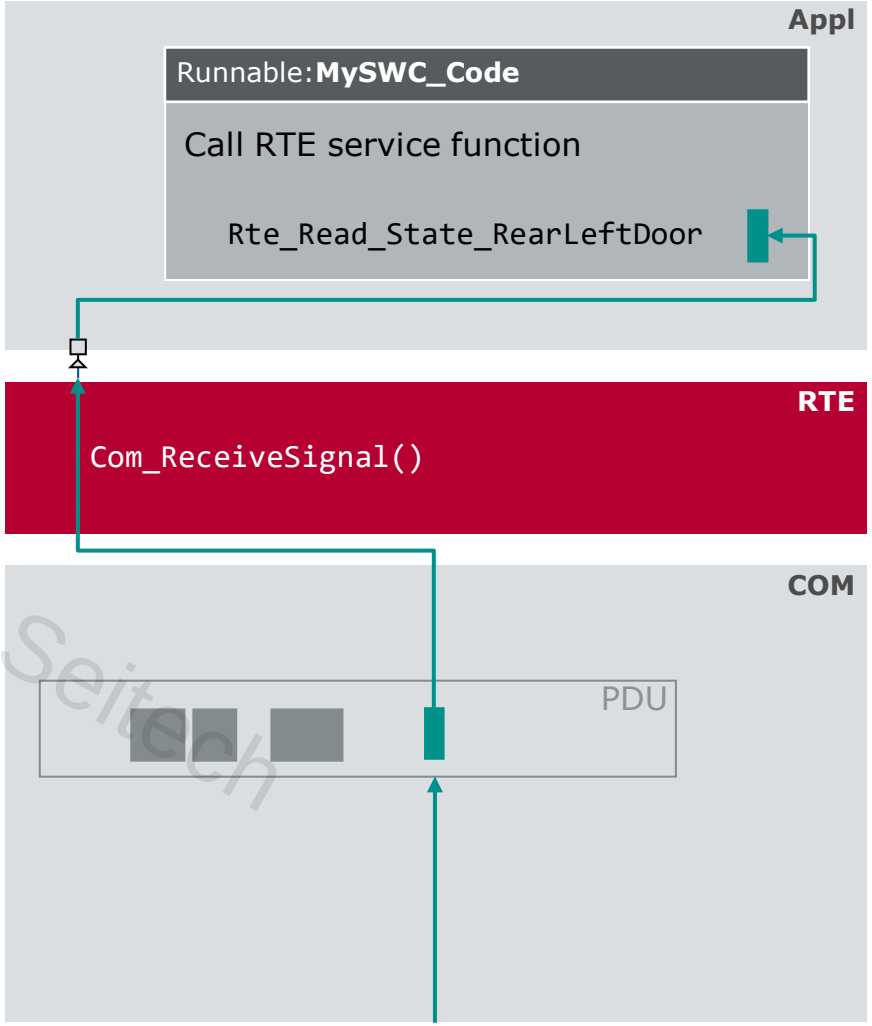
Signal Reception: COM, RTE, Appl

Direct / Explicit Read

- When reading the data element several times, the data may change as the latest signal value received by Com is returned.

Buffered / Implicit Read

- Before the start of the runnable, the signal value is copied to a buffer. The application accesses the data only from the buffer. Therefore the data element remains the same even when the underlying signal value changes.



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- Coming up next

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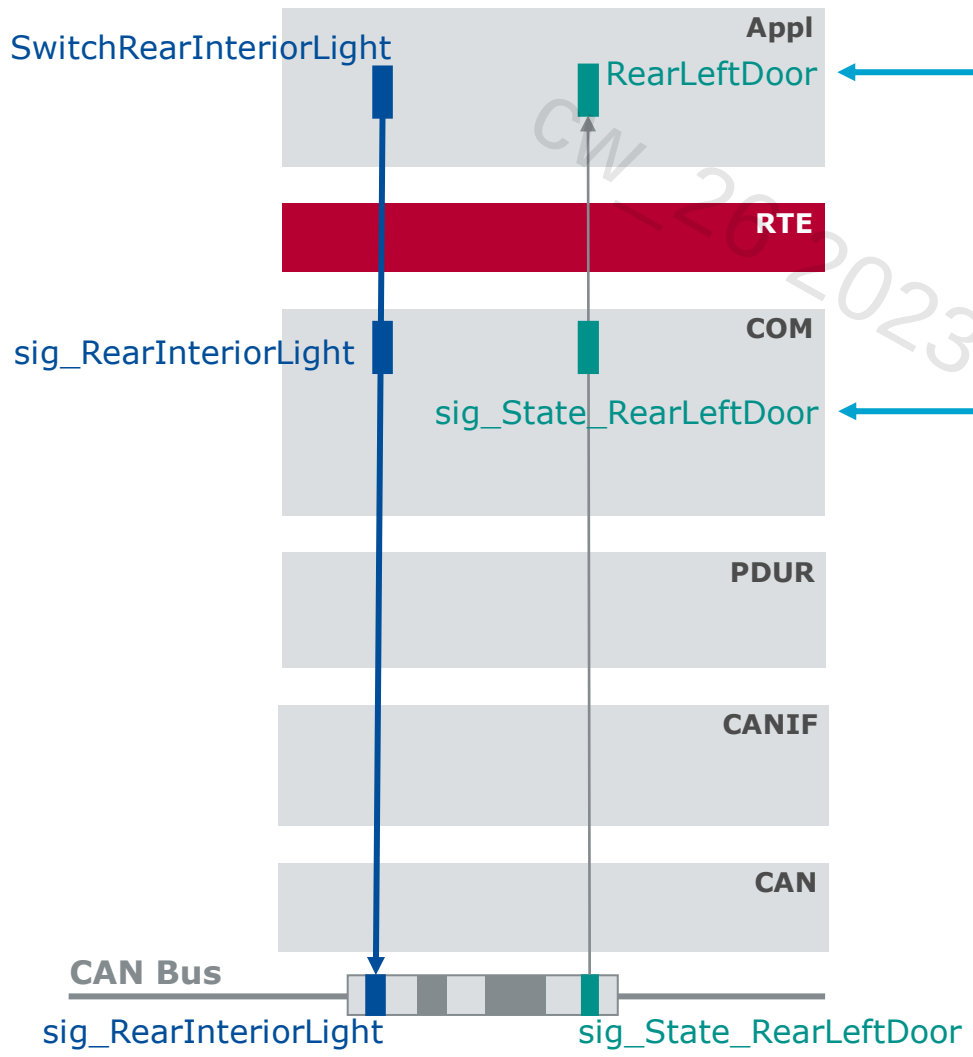
Overview



In DaVinci Configurator

Data Elements have to be assigned to Network (Com) Signals.

This assignment is done when performing the Data Mapping.

- Data Mapping is done for receive and transmit information
- Data Mapping means assigning **Data Elements** to **Network Signals**

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Two ways to do Data Mapping

1. Based on the root element in the hierarchical SW Design

- At the level of the **ECU SW Composition**
  - **Delegation Ports** are required
  - Can be done in both **Developer** and **Configurator**

2. Based on the leaves of the SW Design without respect to the hierarchy

- At the level of **Atomic SWCs**
  - **No Delegation Ports** required
  - This option is only available in **Configurator**

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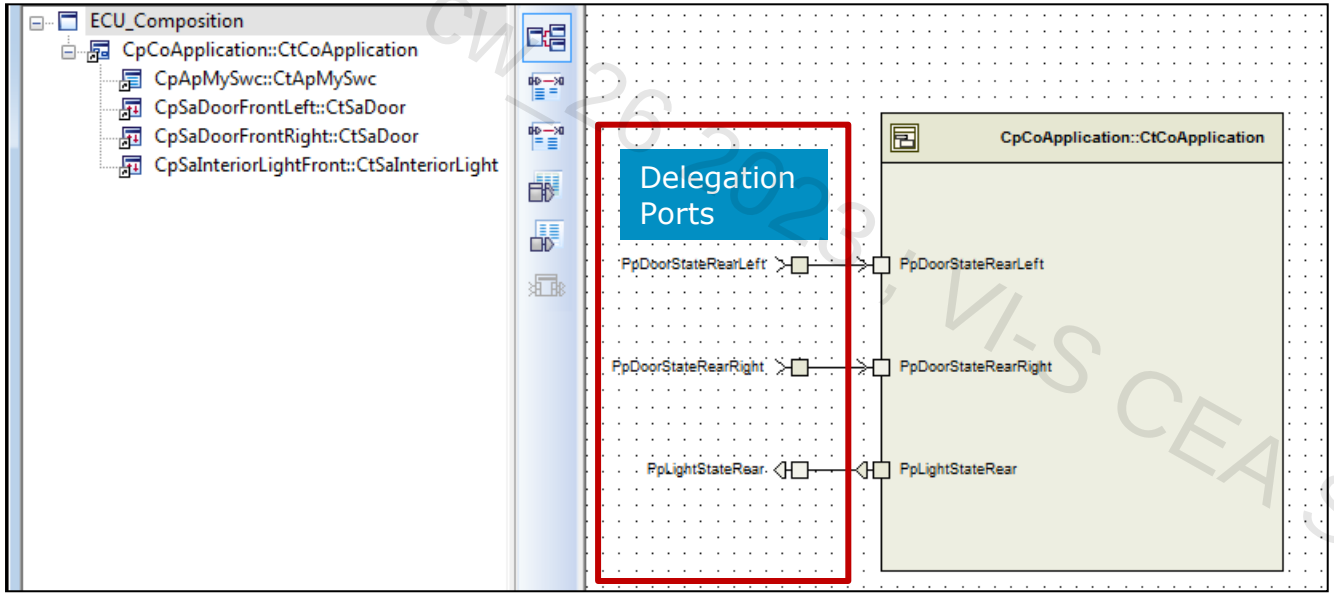
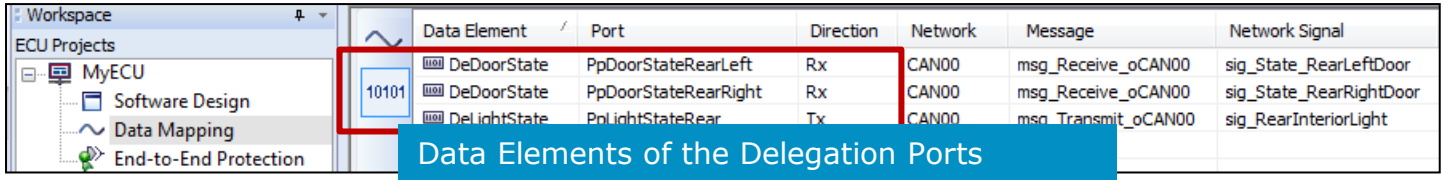
AUTOSAR 3.x style

- **DaVinci Developer** needs Delegation Ports at the ECU Composition level to do Data Mapping.



The **Delegation Ports** promote the Port Prototypes to a higher level in the SWC hierarchy. In the above case this is the top level SWC "ECU_Composition" which is a composition type SWC .



Data Mapping in **DaVinci Developer**: Port Prototypes of the ECU SW Composition

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AUTOSAR 3.x or 4.x style

- In **Configurator**, the ECU SW Composition appears similarly

ECU Software Components

All ▸ ECU Composition ▸ Data Mapping

<Filter>

All
- ECU Composition
  - Port Prototypes
  - Application Connectors
  - Data Mapping

3 of 3 data element prototypes are mapped. Use the Data Mapping Assistant to map the data element prototypes to signals.

Port Prototype	Data Element Prototype	Direction	Channel	PDU	Signal
PpLightStateRear	DeLightState	Tx	CHNL	msg_Transmit_oCAN00	sig_RearInteriorLight
PpDoorStateRearLeft	DeDoorState	Rx	CHNL	msg_Receive_oCAN00	sig_State_RearLeftDoor
PpDoorStateRearRight	DeDoorState	Rx	CHNL	msg_Receive_oCAN00	sig_State_RearRightDoor

Delegation Ports with Data Elements

Data Mapping in **Configurator**: Port Prototypes of the ECU SW Composition

- In addition, Configurator also allows Data Mapping at atomic SWCs further down in the hierarchy. This is usually empty.

ECU Software Components

All ▸ Application Components ▸ CpApMySwc ▸ Data Mapping

<Filter>

- Port Prototypes
- Application Connectors
- Service Connectors
- Task Mapping
- Exclusive Area Impleme
- Data Mapping
- Memory Mapping
- CpSaDisplay

0 of 10 data elements are mapped to signals. Use the Data Mapping Assistant for mapping data elements to signals.

Port Prototype	Data Element Prototype	Direction	Channel	PDU	Signal
PpDisplayState	DeOdometerValue	Tx	<unmapped>	<unmapped>	<unmapped>
PpDisplayState	DeOdometerWriteRequestPending	Tx	<unmapped>	<unmapped>	<unmapped>
PpDoorStateFrontLeft	DeDoorState	Rx	<unmapped>	<unmapped>	<unmapped>
PpDoorStateFrontRight	DeDoorState	Rx	<unmapped>	<unmapped>	<unmapped>
PpDoorStateRearLeft	DeDoorState	Rx	<unmapped>	<unmapped>	<unmapped>
PpDoorStateRearRight	DeDoorState	Rx	<unmapped>	<unmapped>	<unmapped>

Data Mapping in **Configurator:** Ordinary Port Prototypes of an arbitrary atomic SWC can directly be mapped without delegation ports. In this picture the respective data elements are not mapped because this is done on ECU composition level (see figure on top)

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I-PDU Signal Processing

This option determines the call context and the point in time when the signal's

- Rx notification (Rx indication) or
- Tx notification (Tx confirmation) is called.

- IMMEDIATE
  The notification function is called within `Com_RxIndication()` or `Com_TxConfirmation()`. Depending on the lower layer interface, this might be in interrupt context.

- DEFERRED
  The notification function is called on task level during the next call cycle of `Com_MainFunctionRx()` or `Com_MainFunctionTx()`.

Please note that in **DEFERRED** I-PDU Signal Processing, the signal unpacking of the data happens always during the Com_MainFunctionRx, regardless of the setting of IpduSignalProcessing parameter.

That means the user can only access data at earliest after the next Com_MainfunctionRx call.

For **IMMEDIATE** I-PDU Signal Processing the unpacking of the data already happens during the CanIf_RxIndication, i.e., maybe in interrupt context.

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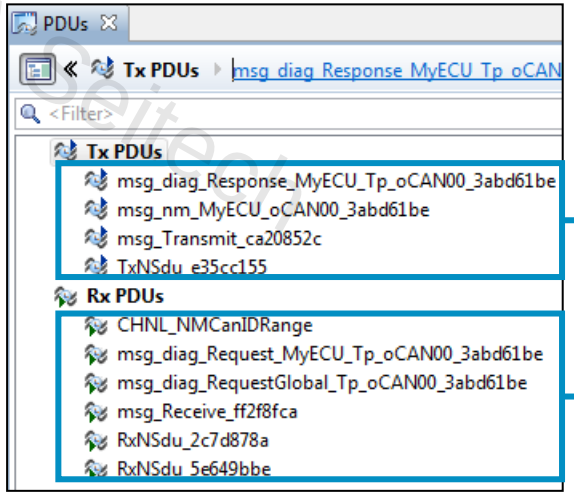
I-PDU Signal Processing

Notification is I-PDU-specific

- Rx and Tx notification functions are defined specifically for a signal
- The kind of notification mechanism (IMMEDIATE or DEFERRED) is specific for an I-PDU

Pdu	Port	Channel	Pdu Groups	Callout	Handle Id	Signal Processing
CHNL_NMCanIDRange	CanIf	CHNL_f276589c	-			
msg_diag_Request_MyECU_Tp_oCAN00_3abd61be	CanIf	CHNL_f276589c	-			
msg_diag_RequestGlobal_Tp_oCAN00_3abd61be	CanIf	CHNL_f276589c	-			
msg_Receive_ff2f8fca	CanIf	CHNL_f276589c	MyECU_oCAN00_Rx_fcef2243		0*	DEFERRED
RxNSdu_2c7d878a	CanTp	<NONE>	-			DEFERRED
RxNSdu_5e649bbe	CanTp	<NONE>	-			IMMEDIATE

Pdu	Port	Channel	Pdu Groups	Callout	Signal Processing
msg_diag_Response_MyECU_Tp_oCAN00_3abd61be	CanIf	CHNL_f276589c	-		
msg_nm_MyECU_oCAN00_3abd61be	CanIf	CHNL_f276589c	-		
msg_Transmit_ca20852c	CanIf	CHNL_f276589c	MyECU_oCAN00_Tx_fcef2243		DEFERRED
TxNSdu_e35cc155	CanTp	<NONE>	-		DEFERRED
					IMMEDIATE



In case update bits have been configured for an Rx signal, the Rx notification is called only if the associated update bit is set

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Mechanism and adjustment of time values

COM can detect timeouts on

- I-PDU level
  - An Rx I-PDU is not received within a predefined timeout time
- Signal level
  - The update bit of a signal has not been set for a predefined timeout time

Timeout time

- Can be configured in DaVinci Developer (SWC) and Configurator (Com)
- Synchronization via Configurator
- The timeout time of an I-PDU is derived from the smallest timeout time of monitored signals within the I-PDU

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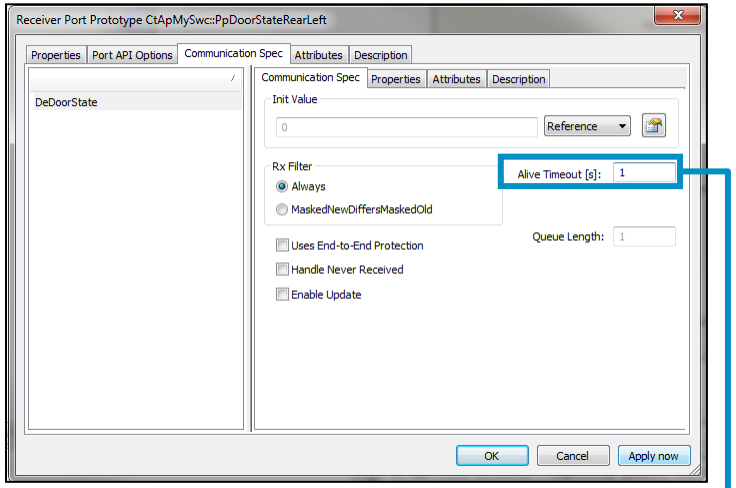
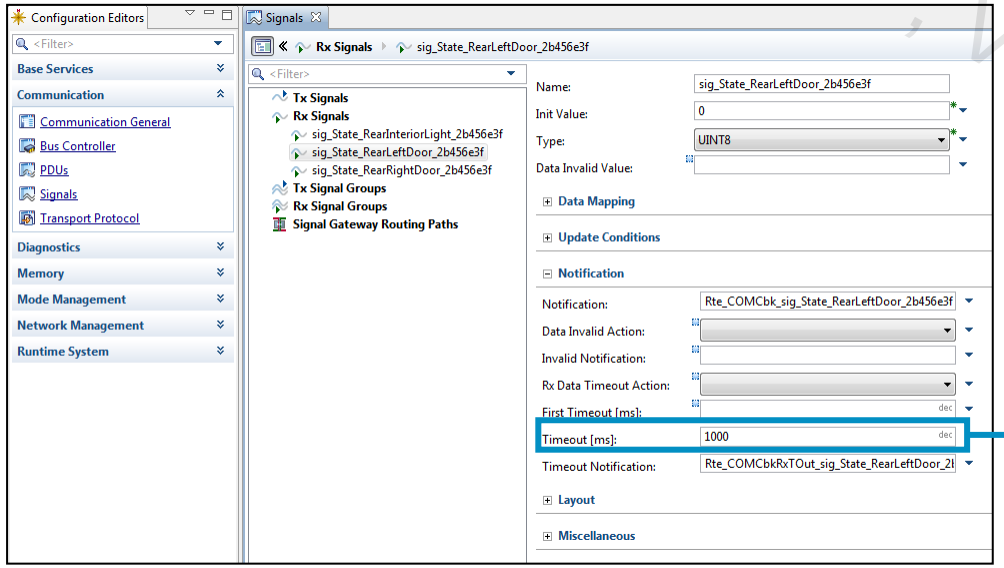
Timeout settings in Configurator and Developer

- Only Inter-ECU timeout monitoring is supported
  - since the Com module provides the Deadline Monitoring functionality



Developer: Alive Timeout in the ComSpec of a data element prototype



Configurator: Related Com signal after synchronization with the Developer project

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Timeout on COM and RTE level

COM Timeout handling:

- A timeout notification is called by Com

  `Rte_COMCbkTOut_<ComSignalName>`

- Optionally, affected signals are set to their initial values (if ComRxDataTimeoutAction is configured to TRUE)

RTE Timeout handling:

- A timeout can trigger a runnable in the RTE
- The return value of the function `Rte_Read_<Pp>_<De>()` indicates the error `RTE_E_MAX_AGE_EXCEEDED`
- In case of implicit S/R communication (`Rte_IRead` API) the status can be determined by the `Rte_IStatus` API

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Timeout on COM and RTE level

- For intra-ECU Sender / Receiver communication, the RTE does not perform timeout monitoring
- RTE can do timeout supervision for asynchronous, intra-ECU Client / Server API calls
- If Com detects a timeout, it will **not** set any Dem event automatically. If a Com timeout shall result in a Dem entry, the application has to set the Dem event by itself.
- Com calls the timeout notification cyclically as long as the timeout condition exists. The cycle time is equal to the configured timeout time.
- In case of a timeout the signal status is set to `RTE_E_MAX_AGE_EXCEEDED`. The signal status can be obtained in the return value of the `Rte_Read_<Pp>_<De>`(…) function call for the respective signal.

→If you set the Alive Timeout value for a data element inside a Port Prototype to a non-zero value, this value will be used in the Com configuration for the signal. The Rte_COMCbkTOut callback will be turned on and the timeout factor will be set to the value from the SWC. This will happen as soon as the ECUEX becomes synchronized in Configurator and the RTE Validation takes place.

→If you have a Com Signal data element relationship of 1:N, then the minimum timeout value of all related data elements will be used as the timeout value.

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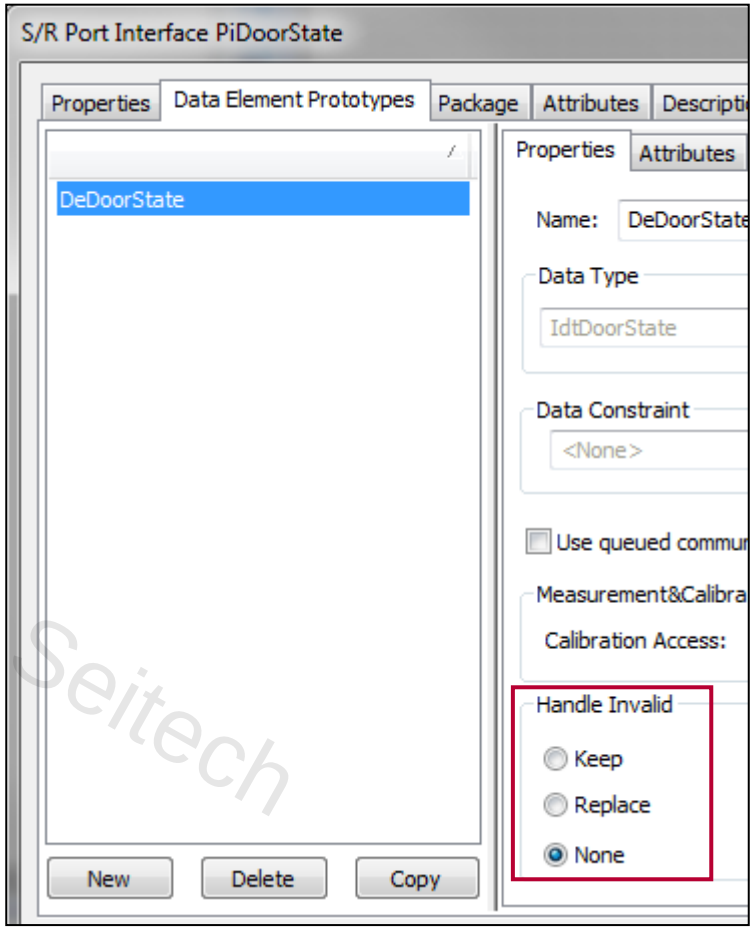
RTE level

- Only for explicit access, non-queued communication, or implicit API
- Inter-ECU and Intra-ECU supported
- Tx Signal (Handle Invalid != none)
  - Send invalid data via
    - `Rte_Invalidate` (explicit API)
    - `Rte_IInvalidate` (implicit API)
- Rx Signal
  - Handle Invalid receive data
    - KEEP (`RTE_E_INVALID` available)
    - REPLACE (no `RTE_E_INVALID` available)
    - NONE (no `RTE_E_INVALID` available)



DEV (ASR4.x): Handle Invalid property is located in the DataElementPrototype inside the **Application Port Interface** definition (ASR3.x has it in the P/R-Port CommSpec)

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Timeout on COM and RTE level

Implicit is buffered and uses Rte_IRead

Explicit is direct and uses Rte_Read

KEEP (`Rte_COMCbkInv_<ComSignalName>`)

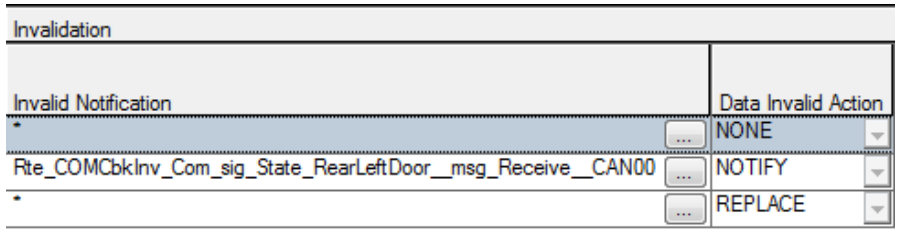
- Keeps the last received valid value in COM
- Std_ReturnType `RTE_E_INVALID`
- Optionally Data Receive Error Event Trigger via the Invalidation Callback

REPLACE (no `Rte_COMCbkInv_<ComSignalName>`)

- Replaces value by "init" value in COM if Com_DataInvalidAction is REPLACE
- No notification to the application that a replacement value is used: Data Receive Event, no `RTE_E_INVALID` available.
- *The Init value is not necessarily from the valid data range!*
- *Data range invalidation (undefined reception range) does not exist in ASR3.x and is a feature of ASR 4.0.*

NONE (no `Rte_COMCbkInv_<ComSignalName>`)

- The invalid COM value propagates to application; no notification of `RTE_E_INVALID`
- The application has to examine the signal value and handle the invalidation on its own



Corresponding GENy settings for the Inter ECU case. KEEP corresponds to NOTIFY, REPLACE is the same like on RTE level.

Remark for the Invalidation intra-ECU (ECU internally, without participation of the Com module):

If the sender data element is configured for invalidation support over "can Invalidate", a built-in check denies the setting "NONE" for the option "Handle Invalid" at the receiver data element.

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COM level

- Precondition for invalidation
  - Only signals which have an enabled *ComSignalDataInvalidValue* can be invalidated at all
- Invalidation of group signals
  1. A **valid** signal group can not contain any **invalid** group signals
  2. An **invalid** signal group contains at least one **invalid** group signal
     - If one of the group signals becomes invalid, this affects the entire signal group

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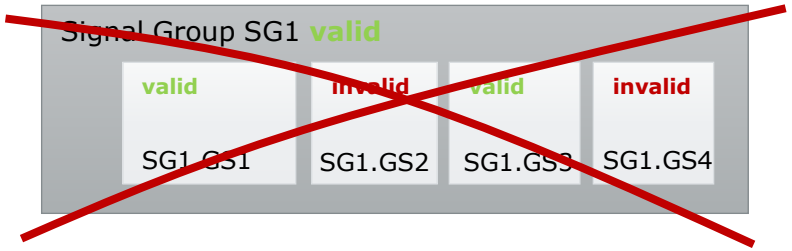
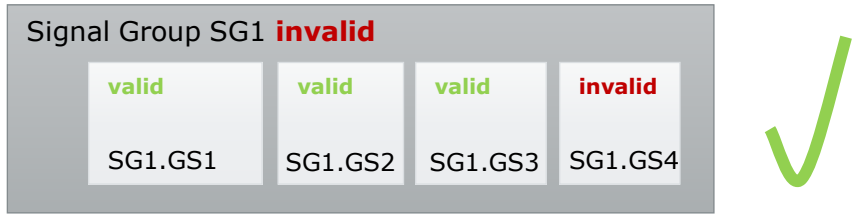
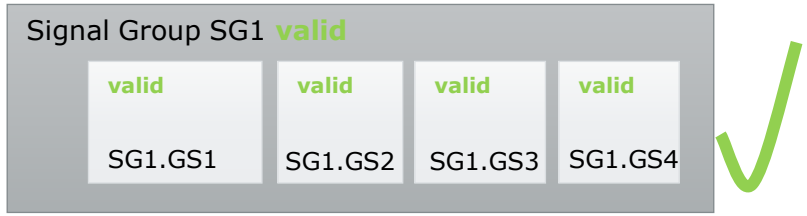
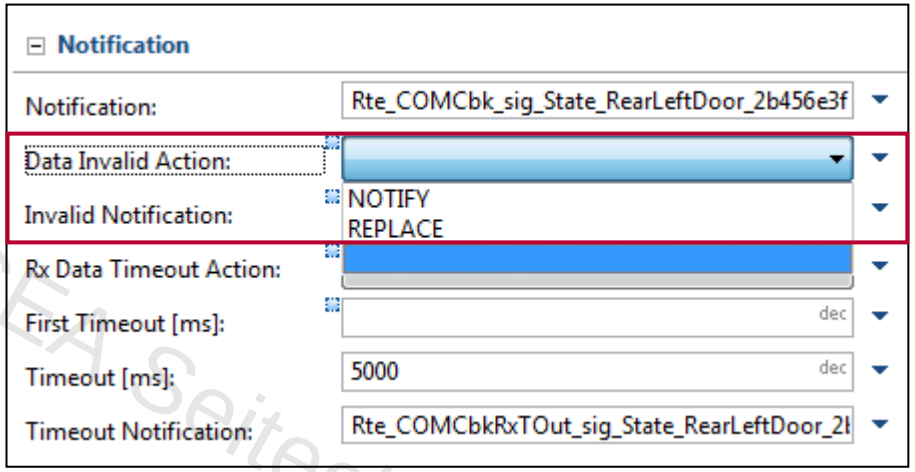






COM level

- Com_DataInvalidAction
  - NONE
    - Nothing happens received invalid value is in the Com buffer
  - NOTIFY
    - The invalid notification `Rte_COMCbkInv_<ComSignalName>` will be called
    - Last valid received value remains in the Com buffer
  - REPLACE
    - Replace the received invalid value in the Com buffer by the Init value



Rx Signal Com_DataInvalidAction

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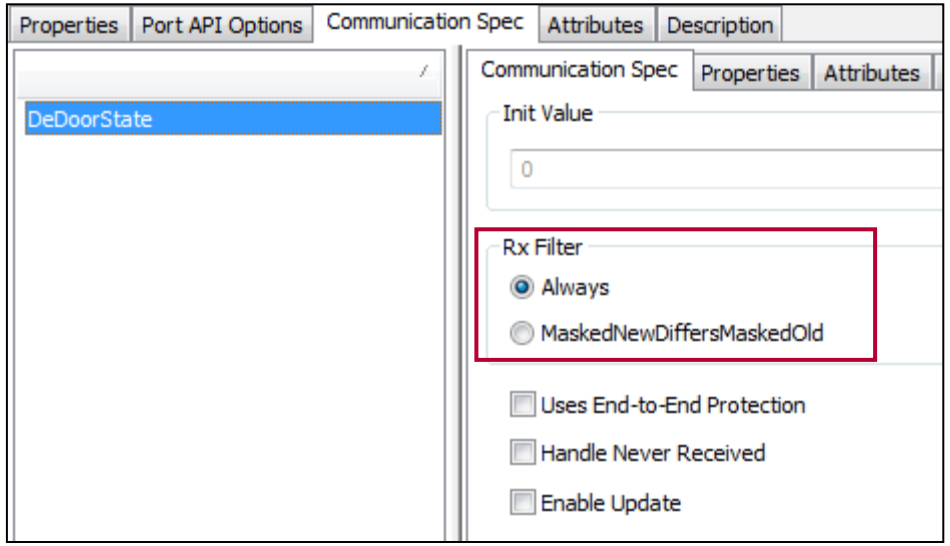
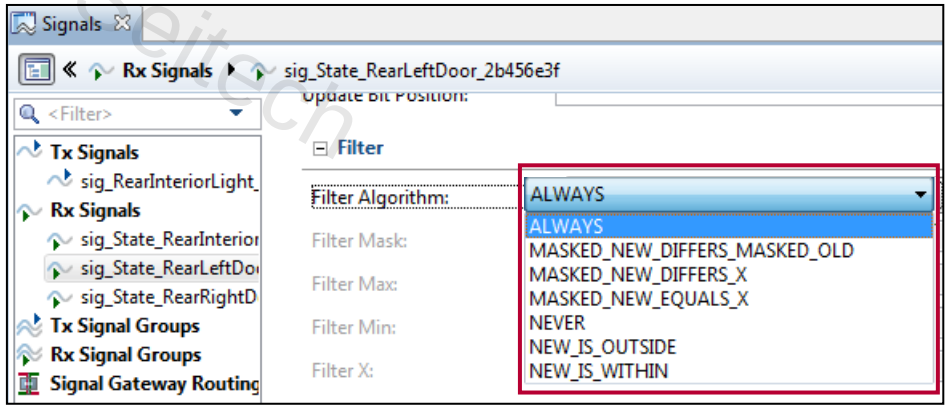
Reception Filtering

- Inter-ECU
- Checkbox in Developer
  - R-Port data element Communication Spec
- Corresponding settings in Com for the Rx Signals
- *Always*
  - every reception causes a notification
- *MaskedNewDiffersMaskedOld*
  - RTE can produce a Data Received Event which only happens “on change”



Developer R-Port Prototype: Rx Filter in the Communication Spec of a Data element



CFG Signal Filter settings

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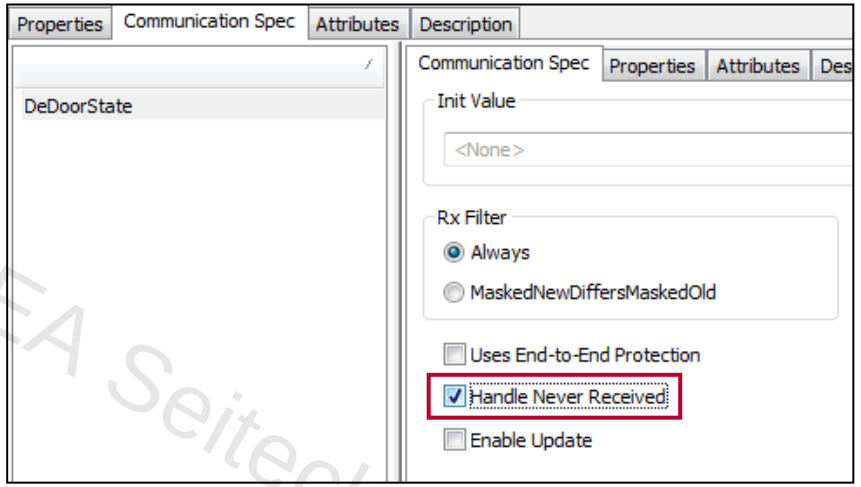
Further Communication Specification settings

- "Handle Never Received" checkbox in Developer
  - R-Port data element Communication Spec
- `Rte_Read` + return Value `Rte_IRead` + `Rte_IStatus`
  - `RTE_E_NEVER_RECEIVED`
  - Since `Rte_Start`, no data received over COM
- `Rte_Read` provides the Init value in the case of `RTE_E_NEVER_RECEIVED`
- Further information
  - In case COM is switched off
    - > `Rte_Write` returns `RTE_E_COM_STOPPED`
  - "Enable Update" checkbox
    - > `Boolean Rte_IsUpdated_<p>_<d>([IN Rte Instance])` tells if new data is received.



Developer R-Port Prototype: Handle Never Received in the Communication Spec of a Data element prototype

This feature is if you need to detect that a signal has been received for the first time by Com.

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Communication

Exercise: Communication

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